Fields of War: Conflict and Climate Vulnerability across the Syrian-Turkish Border*

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Abstract

Conflicts can disrupt agricultural systems, amplifying the toll of war, especially in countries that have limited endowments of land suitable for agriculture. However, because of challenging endogeneity issues, the literature fails to properly estimate the impacts of conflicts on land cultivation. Using a geographic difference-in-discontinuities approach across the Syrian-Turkish border, the study shows that the Syrian Civil War led to cropland abandonment equal to 8.1% of the Syrian area of analysis, alongside decreases in crop greenness of 0.67 standard deviations. These impacts differed in magnitude across years, as the largest reductions occurred in years with the lowest levels of rainfall, with losses in cropland as large as 22% of the area of analysis and reductions in crop greenness of up to 1.6 standard deviations occurring in years of extremely poor precipitation. This points to an increase in climate vulnerability, which likely resulted from the destruction of irrigation infrastructure in Syria.

Keywords: Land Use Dynamics, Cropland Abandonment, Irrigation Destruction
JEL codes: C21, D74, Q15

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1 Introduction

Poor countries heavily reliant on agriculture are among the most exposed to conflict (McGuirk and Burke, 2020). This is a source of vulnerability as conflicts lead to violence and displacement that tend to negatively impact agricultural productivity, causing severe income loss, food insecurity, and malnutrition (Martin-Shields and Stojetz, 2019).

This paper contributes to the literature that focuses on the relationship between conflict, agricultural outcomes, and climate vulnerability by assessing the impact of the Syrian Civil War (2011-2024) on cropland in Syria and neighboring Türkiye. Since the Syrian Civil War broke out, there have been violent events on the Syrian side, leading to, among many other disastrous outcomes, the destruction of irrigation infrastructure and displacement of much of the Syrian population in this area (World Bank, 2022).

Our empirical analysis employs large granular datasets from satellite imagery that classify land use types and measure the intensity of live green vegetation. We use geographic regression discontinuity (GRD) methods to identify the impact of the conflict along the Syria-Türkiye border, alongside a difference-in-discontinuities design—a methodology which combines differences-in-differences analysis with regression discontinuity—to quantify the extent of agricultural losses relative to pre-war differences between Türkiye and Syria. This is made possible by the exogenous border between these countries, drawn by the Ankara Treaty in 1921 along the 37th parallel, without consideration of biophysical characteristics. The analysis shows that agricultural land use declined during the Syrian Civil War, especially in years with poor rainfall, indicating greater vulnerability to drought-like conditions.

Due to security problems, most research on the Syrian Civil War's effects on agricultural production has relied on secondary sources and imperfect measures. The Food and Agriculture Organization of the United Nations (FAO) estimates that fighting in Syria had caused $16 billion of lost agricultural and livestock production in just the first 6 years of the conflict (FAO, 2017). It is notable, for instance, that Syria had been a net exporter of wheat in the prior 20 years (excluding drought years) but the production of this crop fell to 1.5 million tons, approximately 55 percent lower than the pre-conflict average over the 2007-2011 period (FAO and WFP, 2016).

Of the existing literature on climate, agriculture, and conflict, attention tends to focus on how severe climate events contributed to conflict. In an extensive meta-analysis of climate and conflict, Burke et al. (2024) demonstrate that increases in local temperature significantly raise the risk of inter-group and inter-personal conflict. Investigations on Syria specifically have come to similar conclusions, suggesting that the 2006-2011

drought immediately preceding the conflict may have contributed to the subsequent unrest (Gleick, 2014; Kelley et al., 2015).

Comparatively little focus has been paid to how conflict has increased vulnerability to climate shocks, which is the focus of this paper. Cross-country panel studies have shown that fragile and conflict-affected states face larger macroeconomic penalties from drought (Tintchev and Jaramillo, 2024), and household panel evidence from Ethiopia finds that conflict exacerbates the effects of rainfall variability on agricultural input use (Ayalew et al., 2026). These studies, however, face identification challenges common to observational designs: conflict exposure is not randomly assigned, and affected and unaffected areas often differ along many dimensions simultaneously.

Within the Syria context, Li et al. (2022) study the impact of the conflict on cropland across the whole of Syria, measured from satellite imagery from 1998 to 2019. Their research finds significant cropland losses alongside increased interannual variability in cropland extent. Further exploring the mechanisms underlying this wartime agricultural decline, Atik et al. (2024) examine the Idleb and Aleppo provinces of Northwest Syria. Surveying farmers, they find significant decreases in cultivated land area and irrigated area, alongside a significant increase in the use of unregistered small scale groundwater wells, suggesting that weak groundwater governance contributed to poor wartime agricultural outcomes in Syria. While Li et al. (2022) and Atik et al. (2024) study the impact of the Syrian conflict on agricultural climate vulnerability, they do not provide causal identification of this relationship.

Our paper departs from the previous literature in three important ways. First, to our knowledge, and thanks to the GRD and difference-in-discontinuities methodologies that we apply, our paper is the first to provide robust causal evidence of cropland decline attributable to conflict. Second, our analysis assesses both production and productivity effects of the conflict by focusing on the intensive and extensive margins of cropland dynamics. Third, due to the absence of irrigation infrastructure and irrigation governance data, the previous literature could only provide anecdotal evidence of the destruction of irrigation during wars. We fill this gap by proposing a test that indirectly assesses failures in irrigation systems through increased variability of evapotranspiration. Using this test, our findings suggest that failures in irrigation systems contributed to making Syria more vulnerable to climate shocks following the outbreak of conflict.

2 Data and stylized facts

Our investigation focuses on an area comprised of land 50 km north and south of the Syria-Türkiye border, as depicted in the highlighted zone in Figure 1. The regression discontinuity design that we will implement (see Section 3) requires all relevant factors besides treatment (the civil war) to vary smoothly at the boundary (Dell, 2010). For that reason, we focus exclusively on the portion of the border from Hatay Province to the easternmost point along the border with at least 40 km of north/south distance between Syria's border with Türkiye and its border with Iraq. We leave out the side of the Syria-Türkiye border that is closest to the Mediterranean Sea as it differs from the border finalized during the Treaty of Ankara, and does not lend itself to using north/south distance to the border as a running variable. Appendix Figure A1 presents the area of analysis within the larger context of the region for further clarity.

In the rest of the section, we describe the data sources for our geo-spatial data that includes land cover types, a vegetation index, precipitation, and evapotranspiration variables.

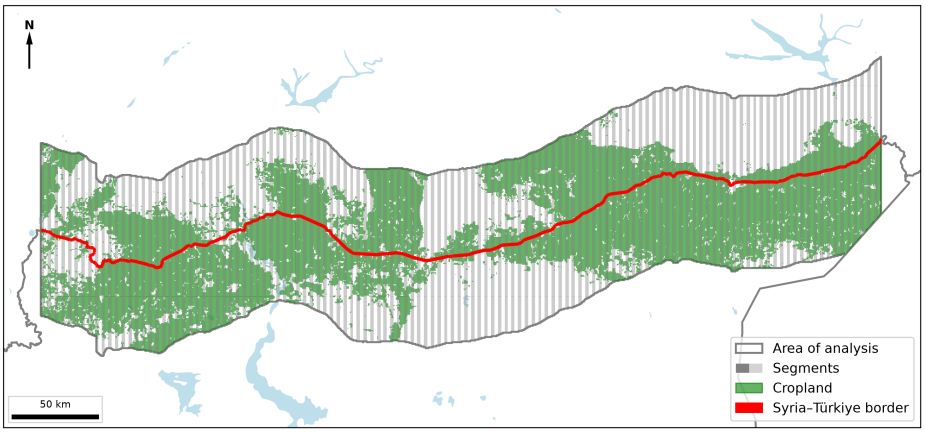


Figure 1: 2010 Cropland with highlighted segments

Sources: MODIS; World Bank Development Data Hub; and Digital Charts of the World. Note: Green pixels represent those which were cropped in 2010. Odd numbered segments are colored with a gray overlay. Cropland data are sourced from MODIS, and country boundaries are sourced from the Official Boundaries shapefiles from the World Bank Development Data Hub. Inland water shapefiles are sourced from Digital Charts of the World.

A. Land cover data

The land cover data is derived using a combination of Moderate Resolution Imaging Spectroradiometer (MODIS) Terra and Aqua reflectance data, which image the entire Earth every 1 to 2 days. The MCD12Q1 product in particular, developed around a

supervised classification of MODIS reflectance, provides global land cover at 500-meter spatial resolution for each year in the 2001-2023 period (Sulla-Menashe and Friedl, 2018). Using the classification provided by MODIS (which classifies a pixel as cropland if at least 60% of its area is cultivated), we converted the land use data into a dummy variable equaling one if a given pixel is identified as cropland, and zero otherwise.

Green pixels in Figure 1 represent pixels which were classified as cropland in 2010, the last year prior to the outbreak of hostilities. An initial exploration of the data reveals land use changes in the study area have been significant in the years following the beginning of the Syrian Civil War. Of the land in Syria within 50 km of the border which was cropland in 2010, 20.0% had transitioned to non-cropland in 2017, compared to just 6.6% of Turkish cropland which had transitioned out of being cropland by 2017. Conversely, of land which was not cropland in Syria in 2010, 2.5% was cropland in 2017, while 7.9% of such Turkish land was cropped in 2017.

B. Vegetation Index

In addition to the land use data, a vegetation index allows us to study cropland losses and gains not only through an extensive, but also from an intensive margin. The intensive margin refers to an increase in the cropland area's intensity of healthy vegetation measured by the Normalized Difference Vegetation Index (NDVI), in which zero means no vegetation and close to one indicates the highest possible density of green leaves (Rouse et al., 1974). These two approaches allow us to study the contraction or expansion of cropland area, as well as the general deterioration or intensification of healthy vegetation.

NDVI is calculated from the visible and near-infrared light reflected by vegetation (Jensen, 2018).[1] Sparse vegetation and unhealthy crops typically reflect most visible red light and yield moderate NDVI values (approximately 0.2 to 0.5), while dense vegetation and crops at the peak of their growth absorb red light and yield high values (approximately 0.6 to 0.9), making NDVI a useful indicator of vegetation health (Brown, 2018). Our NDVI data come from the MODIS MOD13A1 V6 product, a sixteen-day composite at 500m spatial resolution (Didan, 2015). We use the growing-season average (February through April) for each cropland cell.

[1] $\text{NDVI} = \frac{(\text{NIR} - \text{Red})}{(\text{NIR} + \text{Red})}$ where NIR is the reflectance of near-infrared light, and Red is the reflectance of visible red light. As healthy vegetation absorbs most visible red light and reflects most near-infrared light, the light which reflects off such healthy vegetation and into a sensor will include little visible red light and comparatively more near-infrared light, resulting in a high NDVI.

C. Precipitation

Precipitation serves this investigation, both as a check of the continuous nature of independent biophysical variables, as well as an explanatory variable for differences observed in regression estimates. Precipitation data represent the average level of precipitation in millimeters-per-day observed in the growing season (February to April), using data from Climate Hazards Group Infra Red Precipitation (CHIRPS) (version 2). Precipitation data are re-sampled to match the same spatial resolution of our landcover data (at a 500m x 500m resolution).

D. Evapotranspiration

We use evapotranspiration, the sum of water that evaporates and transpires from a given location, to detect the presence of irrigation infrastructure. As with land cover and vegetation, the data come from NASA's MODIS satellite product and represent average values in millimeters per day during the growing season. Evapotranspiration is not used as an agricultural outcome variable, but to explore the mechanisms underlying our results, specifically whether the destruction of irrigation infrastructure led to declines in intensive and extensive margin estimates.

Because irrigation smooths agricultural water consumption in the face of variable precipitation, evapotranspiration that is smoothed between periods of high and low precipitation can indicate functioning irrigation infrastructure. Such satellite proxies are particularly useful where direct measures are unavailable. Ideally, we would restrict the sample to pixels that had irrigation infrastructure when the conflict erupted. Because such measures are unavailable, we instead use a proxy sample of irrigated land: pixels that were cropped in at least 8 of the 10 pre-conflict years (2001-2010).

Table A1 in the appendix presents summary statistics (averages over the pixels of our study area on each side of the border) for the four above variables and for all the years of the study period.

Appendix Figure A2 plots the four variables over the study period, averaged within 50 km of the border and indexed so that the 2001-2004 average equals 100.[2] Before the war, cropland and NDVI move in tandem in Syria and Türkiye. After 2011 they diverge, with Syria showing greater year-to-year volatility, and evapotranspiration follows a similar pattern. Precipitation, by contrast, follows similar paths in both countries throughout, suggesting that the post-2011 divergence does not simply reflect differences

[2]We index to the 2001-2004 average rather than to 2001 alone because 2001 had the second-highest precipitation of the 23 years analyzed, which inflated agricultural indicators.

in rainfall. We explore these patterns in the causal analysis in Section 4.

3 Methodology

Figure 1 shows the portion of the border, about 500 km in length, along which we conduct our spatial regression discontinuity analysis. As explained in Section 2, we exclude the westernmost portion of the border along the Turkish province of Hatay, as well as portions of the eastern border without sufficient bandwidth in Syria. A bandwidth of 50 km is used throughout the investigation.[3] The gray and white stripes in Figure 1 indicate vertical segments which are used in regression estimates to control for potential east-west differences in geographic conditions. Segments are defined by subdividing the border into 250 slices of equal width, each about 2 km wide and each defined by a range of latitude values.

Previous studies have relied on a geographic regression discontinuity (GRD) design that exploits the natural experimental environment inherent to geography to achieve causal identification. The rationale is that spatial units on either side of a geographic boundary are plausibly similar, and thus constitute a genuine counterfactual for one another. The continuity of covariates relevant to agricultural outcomes is supported by the history of the border: because it was drawn by European powers without regard to geographic features such as rivers or mountains, it is unlikely to coincide with discontinuities in agroecological conditions.

$$y_i = \alpha_0 + \beta_1 Syr_i + \alpha_1 d_i + \alpha_2 Syr_i d_i + \alpha_3 d_i^2 + \alpha_4 Syr_i d_i^2 + \gamma_i + \varepsilon_i \tag{1}$$

Formally, we use the above regression model to estimate the war's impact on relevant indicators, where y_i is the outcome of interest for pixel i (cropland dummy or NDVI for the discontinuity analysis, or precipitation to check continuity); Syr_i is a dummy variable, equal to one for pixels within Syria, and zero for those in Türkiye; d_i is the north-south distance between pixel i and the Syria-Türkiye border; γ_i are segment fixed effects; and ε_i is the idiosyncratic error term. Note that by construction, d_i is negative for pixels within Turkish territory, and positive for within Syria. β_1 represents our estimate of the treatment effect, quantifying the discontinuous jump in outcome y between Syria and Türkiye. Standard errors are clustered by segment.

[3]This bandwidth uses most data with comparable underlying agroecological conditions (i.e. it avoids including asymmetric inclusion of watersheds from the Tigris river in Türkiye and from Lake Assad in Syria). We prefer to use our larger agroecologically-consistent bandwidth of 50 km based on comparability of geographic features rather than on an aspatial year-specific statistical optimization.

Regression estimates using Equation 1 are calculated independently for each year within the scope of this investigation, both prior to and during the conflict, with null results in pre-war years serving to validate statistically significant spatial discontinuities which emerge in wartime estimates.

This methodology follows a popular precedent of using spatial regression discontinuity analysis in cases where borders were drawn by colonial powers without consideration of local geographic or cultural factors (Dell, 2010; Michalopoulos and Papaioannou, 2014; Guarnieri and Rainer, 2021). In specifying our regression equation, we follow Gelman and Imbens (2019), who recommend quadratic and local linear regression equations over higher-order polynomials, choosing a quadratic specification with linear robustness checks. This choice is further supported by Lee and Lemieux (2010), who recommend higher-order polynomials for wider bandwidths, which allow for us to take maximal use of rich satellite data.

What differentiates this investigation from most others which use spatial regression discontinuity is the use of time series data to show the emergence of a discontinuity over time. To exploit this time dimension, we additionally chose to combine our GRD equation with a difference-in-difference strategy, utilizing a difference-in-discontinuities methodology outlined in Equation 2. Here, the first difference is our calculated discontinuity estimate along the Syria-Türkiye border, while the second difference is that between pre-war and wartime outcomes.

$$\begin{aligned} y_{i,t} = \alpha_0 + \beta_1 Syr_i + \beta_2 Syr_i War_t + \sum_{t='01}^{'23} \alpha_{1t} Year_t d_i + \sum_{t='01}^{'23} \alpha_{2t} Year_t Syr_i d_i \\ + \sum_{t='01}^{'23} \alpha_{3t} Year_t d_i^2 + \sum_{t='01}^{'23} \alpha_{4t} Year_t Syr_i d_i^2 + \sigma_{i,t} + \varepsilon_{i,t} \end{aligned} \tag{2}$$

In this equation, War_t is a dummy variable equal to zero for years before 2011, and one for years after; 2011 is omitted from pooled regression specifications, as hostilities broke out during the agricultural cycle in 2011, such that 2011 does not fit cleanly into either categorization. $Year_t$ are a series of year-specific dummy variables. σ_{it} are segment-year fixed effects. β_1 estimates the extent of the potential discontinuity in y prior to 2011, while β_2 is analogous to a difference-in-difference estimator, estimating the difference in spatial discontinuities in y between wartime and pre-war years. Standard errors in pooled regressions are clustered by segment-year.

This regression specification mirrors other investigations which utilize difference-in-discontinuities methodology (Grembi et al., 2016; Picchetti et al., 2026). Where this investigation has less precedent is in use of difference-in-discontinuities analysis in a

spatial context. The clearest analogous investigation is the working paper Ferguson and Kim (2026), which exploits both temporal and spatial differences in exposure to a policy introduction to understand its effects on crop greenness.

Equation 2 assumes that any difference in the spatial discontinuities in outcome variables induced by the onset of the civil war would result in a level shift in the magnitude of the discontinuity (as captured by β_2). Relaxing this assumption somewhat, we also modify the difference-in-discontinuities specification to explore whether the discontinuities in outcomes are sensitive to precipitation (see Equation 3). Here, P_t is the average precipitation within 20 km of the border during the agricultural cycle (February to April) in year t.[4] A positive β_3 would indicate that the discontinuity in y was more positive (outcomes were better in Syria relative to Türkiye) during years of good precipitation relative to years of poor precipitation, or inversely, that drought-like conditions induce a negative shift in our spatial regression discontinuity estimate. This would potentially capture differences across Syria and Turkiye in their abilities to face droughts, likely thanks to differences in their irrigation system. Similarly, a positive β_4 would indicate that such a difference in susceptibility to poor precipitation between Syria and Türkiye was larger in wartime years relative to the pre-war period. This would likely capture a deterioration of the irrigation system in Syria relative to Türkiye due to the war.

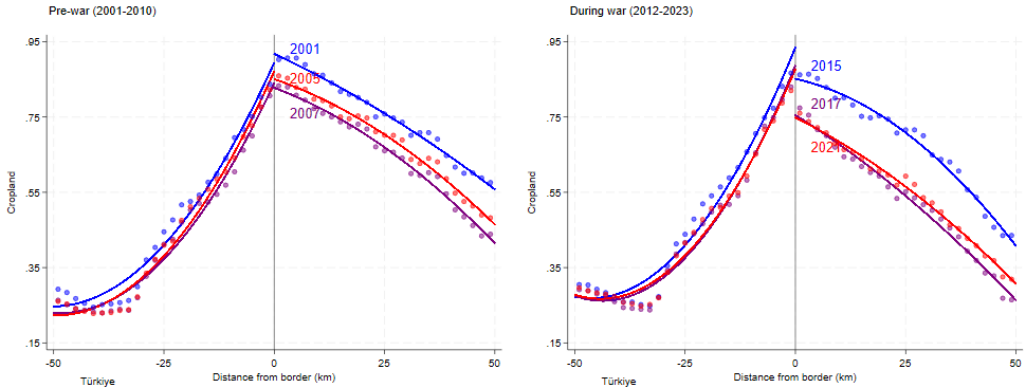
$$\begin{aligned} y_{i,t} = \alpha_0 + \beta_1 Syr_i + \beta_2 Syr_i War_t + \beta_3 Syr_i P_t + \beta_4 Syr_i War_t P_t + \sum_{t='01}^{'23} \alpha_{2t} Year_t d_i \\ + \sum_{t='01}^{'23} \alpha_{3t} Year_t Syr_i d_i + \sum_{t='01}^{'23} \alpha_{4t} Year_t d_i^2 + \sum_{t='01}^{'23} \alpha_{5t} Year_t Syr_i d_i^2 + \sigma_{i,t} + \varepsilon_{i,t} \end{aligned} \tag{3}$$

4 Analysis

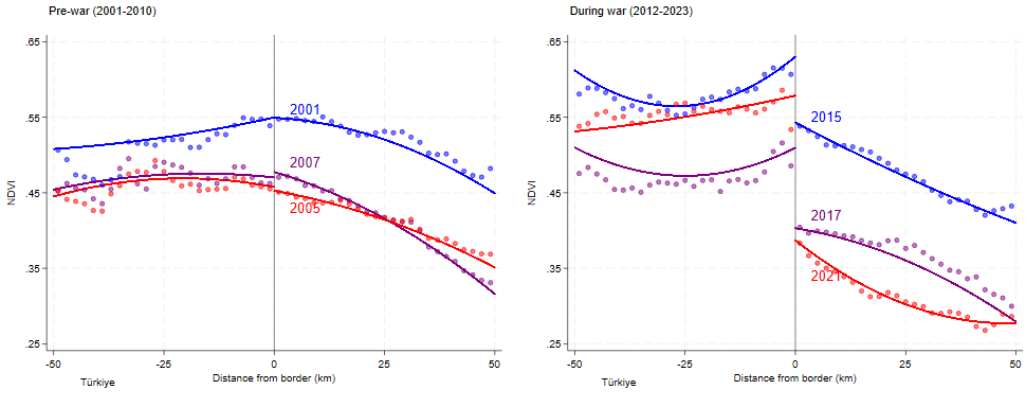
A. Main Results

Panel A of Figure 2 illustrates principal regression estimates for the cropland indicator. Here, six years, chosen to span periods of high, moderate, and low rainfall (respectively in blue, purple, and red) in both pre-war and wartime years (respectively on the left and on the right), are plotted alongside each other. Graphed

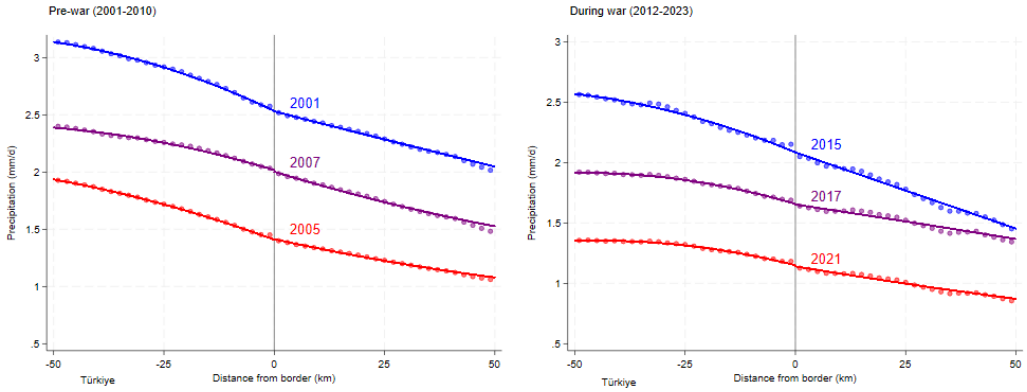
[4] We chose to measure precipitation within 20 km of the border to capture potential variation across the border at a scale consistent with that of the raw precipitation data (5 km x 5 km), affording about four different precipitation measures north and south of the border.



(a) Cropland



(b) NDVI



(c) Precipitation

Figure 2: Regression fit lines by distance to the border

Note: Horizontal axis values indicate distance to the border (negative values for Türkiye and positive values for Syria). Lines are colored based on average annual precipitation, with blue lines experiencing high precipitation, purple lines experiencing moderate, and red lines experiencing low precipitation. Plotted lines represent regression estimates while dots represent averages by distance by 2 km bins. For NDVI, because non-cropland pixels are excluded from analysis, the set of pixels contributing to each distance bin's average may vary in its east-west composition. As a result, the unadjusted bin averages shown in the scatterplot may differ slightly from the regression fit lines, which control for east-west variation through segment fixed effects.

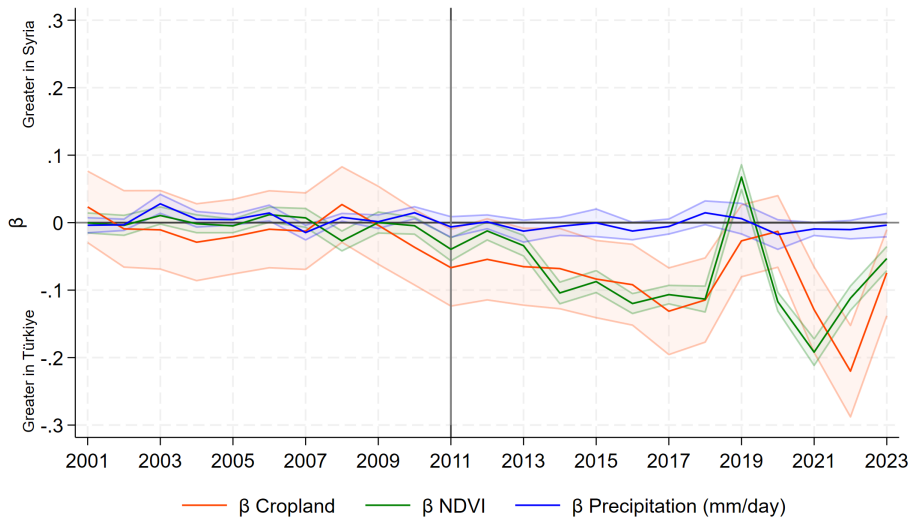


Figure 3: Main regression estimates by year

Note: Regression results represent estimates of the GRD discontinuity coefficient (β_1) using Equation 1. Error bars represent 95 percent confidence intervals, clustering standard errors by vertical segment. Complete regression tables for cropland and NDVI can be found in the appendix in Tables A2 and A3 respectively.

lines represent the regression estimates of land use, using Equation 1, while colored dots provide averages of the underlying data. Discontinuity estimates—β_1 in Equation 1—are represented by the vertical distance between right-hand and left-hand limits of regression fit lines as they approach the cutoff at zero kilometers. Precise regression estimates can be found in Table A2 in the appendix.

Here, while discontinuities are absent in each pre-war year, all three years during the war show significant negative discontinuities. Additionally, it's notable that the extent of the discontinuity in wartime years is not constant; specifically, the largest discontinuities can be observed in years with the poorest rainfall (red lines).

Panel B of Figure 2 mirrors the same analysis, but for the intensive margin estimates using NDVI. As with land use, whereas pre-war regressions indicate no statistically significant discontinuity along the border, large negative discontinuities emerge during wartime years. Just as in the analysis along the extensive margin, the largest discontinuities emerge in years with the lowest levels of rainfall, with relatively small discontinuities, if any at all, in wartime years with high levels of precipitation.

Panel C of Figure 2 replicates a similar analysis with precipitation during the agricultural period (February to April) as the relevant outcome variable. This figure serves to mitigate concerns that discontinuities in land use and NDVI might emerge as a result of differences in rainfall between the two countries. Rather, precipitation consistently varied continuously across the border in both pre-war and wartime years.

Whereas Figure 2 shows select years' regression estimates, Figure 3 represents discontinuity coefficient estimates for all years between 2001 and 2023 along the same three indicators.

For cropland (red line), the estimated regression discontinuity coefficient for each year pre-war is not statistically distinguishable from zero at any conventional level, while every wartime estimate, aside from two notable exceptions, is negative and statistically significant.

A similar trend emerges for NDVI (green line). Of the eleven pre-war years included in analysis, nine are indistinguishable from zero at conventional significance levels, while two outlier years emerge with statistically significant results; one positive (2006) and one negative (2008). Despite this, the magnitude of the effect suggested by the coefficient estimates is relatively small in these outlier pre-war estimates. After the onset of the war, significant negative discontinuities emerge for nearly all years, almost all of which are larger in magnitude than the outlier pre-war estimates.

It's notable, however, that, for cropland in 2019 and 2020, no significant discontinuity is estimated. Additionally, we obtain a positive discontinuity estimate for NDVI in 2019, suggesting higher outcomes in Syria compared to Türkiye for that year. This effect is temporary, as discontinuities reemerge for both outcome variables by 2021. The factor which seems to best explain these outlier years is their exceptionally high levels of rainfall. Average rainfall within 20 km of the border in 2019 was 70% greater than the average rainfall in all other years between 2001 and 2023, and 19% greater than in 2001, the year with the next highest precipitation. 2020 also ranked as having the third highest precipitation in the 23 years of available data.[5]

Our broad result of zero pre-war discontinuity along both extensive and intensive margins, followed by significant negative wartime discontinuities demonstrated in the series of single-year results (Figure 3; Table 1), is consistent with the difference-in-discontinuities model specified in Equation 2, as shown in Columns 1 and 3 of Table 1. These results show, for both cropland and NDVI, no significant prewar discontinuity, while, during the war, a negative discontinuity emerges, suggestive of a relative reduction in cropland equal to 8.1 percent of land area, as well as a reduction in NDVI of 0.079, both of which are statistically significant at the 1 percent level. For context, between 2001 and 2010, NDVI of Syrian cropland within the area of analysis averaged 0.44, with a standard deviation of 0.12, suggesting a 0.67 standard deviation decrease in crop greenness after the onset of the civil war.

[5]The average level of rainfall by year for the entire 50 km bandwidth is also available in appendix Table A1, which outlines summary statistics.

Table 1: Pooled Regression Results

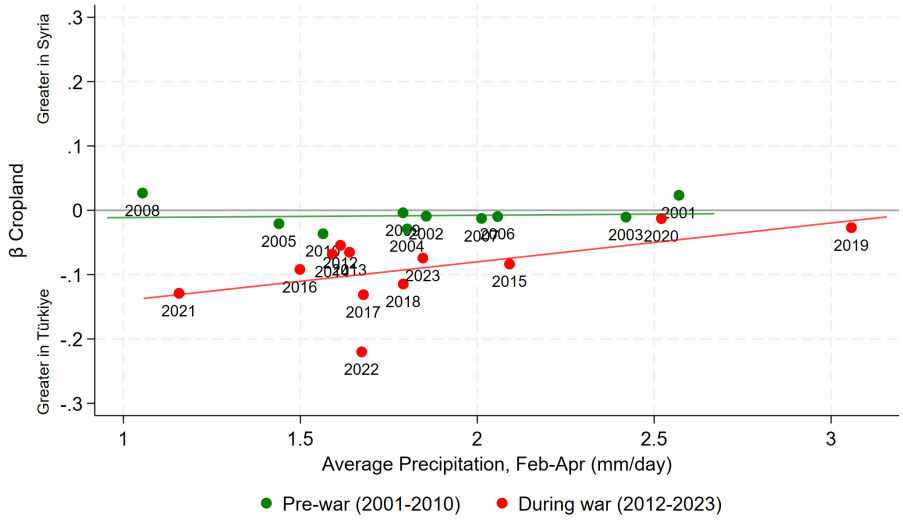
	Cropland		NDVI		ET	
	(1)	(2)	(3)	(4)	(5)	(6)
Syria	-0.008	-0.015	-0.001	-0.034***	0.000	-0.219***
	(0.009)	(0.040)	(0.002)	(0.010)	(0.007)	(0.034)
Syria x Precip.		0.004		0.018***		0.118***
		(0.021)		(0.005)		(0.018)
Syria x Wartime	-0.081***	-0.186***	-0.079***	-0.202***	-0.305***	-0.575***
	(0.013)	(0.052)	(0.003)	(0.015)	(0.011)	(0.046)
Syria x Wartime x Precip.		0.057**		0.066***		0.147***
		(0.027)		(0.008)		(0.023)
Observations	5,389,076	5,389,076	2,846,545	2,846,545	2,730,068	2,730,068
R^2	0.276	0.276	0.529	0.531	0.597	0.599

Note: All years between 2001 and 2023 (except 2011) are included in the analysis. Wartime is a dummy for whether a given year is post-2011, Precip. refers to the average precipitation of all pixels within 20 km of the Syria-Türkiye border in year t. Columns 1, 3, and 5 were estimated using Equation 2, while columns 2, 4, and 6 use Equation 3. ET refers to Evapotranspiration. The coefficients *Syria*, *Syria x Precip.*, *Syria x Wartime*, and *Syria x Wartime x Precip.* correspond to β_1, β_3, β_2, and β_4 respectively in Equations 2 and 3. All regressions were estimated using $year$ x $segment$ fixed effects.

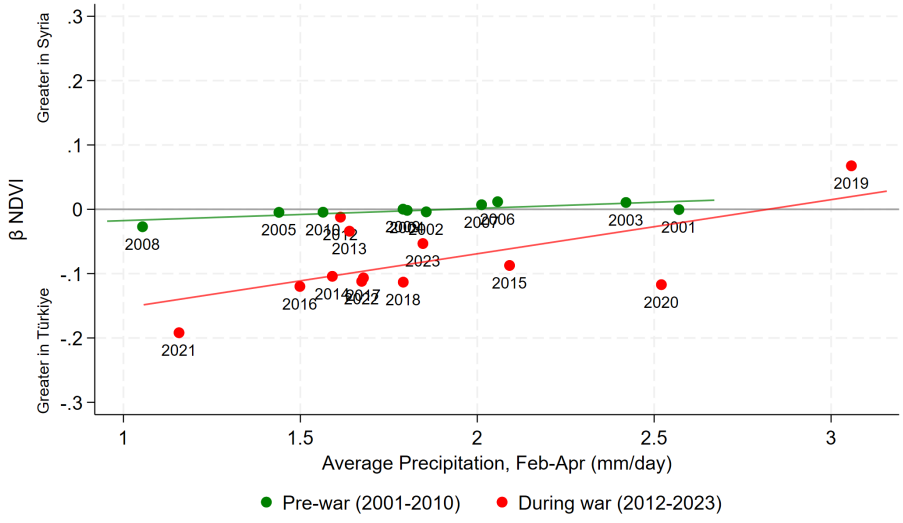
B. Potential Mechanisms

In our analysis, the two years which showed no significant negative discontinuities in our key variables (2019 and 2020, see Figure 3) occurred during years of extremely high precipitation. Likewise, the years with the largest discontinuities were those with extremely low levels of rainfall. Considering irrigation's ability to smooth agricultural water consumption across time, it seems likely that growing discontinuities during years of poor precipitation could be the result of inferior irrigation systems in Syria relative to Türkiye during wartime years, likely due to the wartime degradation and destruction of infrastructure which had previously enabled such smoothing in water use over time in Syria. Satellite imagery in Appendix Figure A3 shows a clear example of destruction in irrigated cropland in the Al Hasakah Governorate, Syria, while Turkish cropland on the other side of the border continued its irrigation.

Figure 4 further investigates the relationship between levels of precipitation and relative differences in outcomes between Syria and Türkiye. For each panel (Panel A for cropland, and Panel B for NDVI), along their vertical axes, figures represent the annual values of β_1 from Equation 1—our discontinuity coefficient estimate shown in Figure 3 and the *Syria* coefficient in Appendix Tables A2 (Cropland) and A3 (NDVI). The horizontal axis represents the average precipitation during the crop growing season. Least-squares fit lines represent the pre-war (green) and wartime (red) relationships between precipitation and regression discontinuity estimates. In Panel A, the flat



(a) Cropland



(b) NDVI

Figure 4: Discontinuity estimates by average annual precipitation

Note: Vertical axis values represent estimates of GRD discontinuity coefficients ($beta_1$) using Equation 1, as available in Tables A2 and A3. Horizontal axis values represent average precipitation within 20 km of the Syria-Türkiye border in year t. Fit lines are linear best fit lines by time period. 2011 cannot be assigned to either pre-war or during-war status because the conflict started during the year's cropping season and is therefore omitted.

near-zero pre-war best-fit line (in green) indicates that precipitation and discontinuities in land use were uncorrelated prior to the conflict. Whereas this is somewhat expected given the null results pre-war, the fact that even marginal pre-war regression estimates are uncorrelated with precipitation indicates that precipitation and discontinuities in land use were uncorrelated prior to the conflict. After the onset of the war, however, a different picture emerges. The largest negative wartime discontinuities were observed during years of poor precipitation, near-zero estimates emerged during years of good precipitation, and the relationship between the two variables was positively sloped across years.

Likewise, Panel B of Figure 4 shows similar findings for intensive margin results. Specifically, while pre-war discontinuities in NDVI are minimal and largely insensitive to changes in precipitation, effect estimates during the war largely fall below pre-war estimates in years of similar precipitation. Additionally, the magnitudes of such negative discontinuities grow as precipitation weakens, as demonstrated by the upward sloping red linear fit line. The positive 2019 discontinuity for NDVI is consistent with less smoothing of groundwater use due to weaker water management governance (Atik et al., 2024) leading to excessive use of water in Syria in the context of extraordinary rainfall.

Such a relationship between the magnitude of regression discontinuity estimates along intensive and extensive margins results alike and the average level of precipitation is further explored using Equation 3—the difference-in-discontinuities estimate which includes a precipitation interaction term—in Columns 2 and 4 of Table 1 for cropland and NDVI respectively. The regression results in Column 2 suggest that, pre-war, across all levels of precipitation, spatial discontinuities in cropland were indistinguishable from zero. During the conflict, however, the positive and statistically significant coefficient estimate for the interaction between our Syria dummy variable and precipitation suggests that, for an increase in average precipitation of 1 mm/day, the discontinuity in cropland would shrink by 5.7 percentage points, or inversely, that conditions of poor rainfall further exacerbated wartime decreases in agricultural land-use. For NDVI, whereas a small positive coefficient for the interaction term between our Syria dummy and precipitation is observed pre-war, that interaction term grew by 0.066 per mm/day of precipitation during the war, supporting the finding that wartime declines in vegetation were largest in years of poor precipitation. The small positive yet statistically significant interaction coefficient in Column 4 of Table 1 suggests that, pre-war, water governance in Syria was weaker than in Türkiye, making Syrian cropland NDVI slightly more sensitive to changes in precipitation. This is captured by the small positive slope of the green pre-war fitted line in Panel B of Figure 4.

One possible explanation of worsening discontinuities in years of poor precipitation is the wartime destruction of irrigation infrastructure. Because no direct measure of irrigation infrastructure is available, we use the variation in evapotranspiration as a proxy for its presence. Our assumption is that, in the presence of functioning irrigation systems, evapotranspiration is smoothed over time, leading to minimal differences between years of high and low precipitation. On the other hand, highly variable evapotranspiration in the face of variable precipitation may indicate the absence or poor quality of irrigation infrastructure.

Figure A4 in the appendix mirrors the analysis done in Figure 4, but using evapotranspiration as the relevant indicator. Vertical axis values represent discontinuity estimates in evapotranspiration using Equation 1. Again, horizontal axis values represent the average precipitation in the relevant year. This analysis shows that, irrespective of levels of precipitation, Syria and Türkiye saw similar levels of evapotranspiration along the border prior to the war, as discontinuity estimates were small and centered around zero. After the war began, however, large negative discontinuities emerge, with a strong positive relationship between the size of the discontinuity and the level of precipitation. This relationship is also reflected in Column 6 of Table 1, as the interaction term between Syria and precipitation more than doubled during wartime years.

This sensitivity to years of poor rainfall is suggestive of destruction of irrigation infrastructure, which—when functioning well—is able to smooth water use over time, such that large differences in rainfall between years do not result in large differences in agricultural water use (or large differences in evapotranspiration).

Our finding of increased precipitation sensitivity during wartime is consistent with the interannual variability documented in Li et al. (2022), who observe that, within their period of analysis (1998-2019), the largest cropland losses and gains occurred in 2016 and 2019 respectively. As shown in Appendix Table A1, of the wartime years within the period of analysis of Li et al. (2022), these same years represent those with the lowest and highest levels of wartime precipitation respectively in Northern Syria.

5 Robustness Checks

We conducted a series of checks, described below and shown in Appendix Figures A5 through A15. Figure A5 compares our principal year-specific GRD results against these checks, Figures A6 through A11 replicate the analysis at alternative bandwidths and alternative regression specifications, and Figures A12 through A15 do the same for

the difference-in-discontinuities models.

A. Year-specific regressions

Linear model. The quadratic form chosen in our principal analysis allows for curvature in the relationship between distance and outcomes. As an alternative, we use a linear form with triangular statistical weights, which downweight observations farther from the cutoff. Equation 4 below is the same as Equation 1 except with 2nd order terms omitted, and uses a 25 km bandwidth with weights equal to $\frac{25\text{km}-|d|}{25\text{km}}$, where d is the vertical distance to the Syria-Türkiye border. This specification is consistent with Lee and Lemieux (2010), who recommend triangular weights and shorter bandwidths for linear specifications.

$$y_i = \alpha_0 + \beta_1 Syr_i + \alpha_1 d_i + \alpha_2 Syr_i d_i + \gamma_i + \varepsilon_i \tag{4}$$

Segment fixed effects and omitted 5 km swath. In place of the 250 segments, each about 2 km wide, used in the principal model (Figure 1), we perform checks using 100 segments of about 5 km, and segments as narrow as the 500m resolution of the data. We also estimate a "donut" model, which guards against outliers near the cutoff by omitting data points within 2.5 km of the border in either direction.

Alternative bandwidths. To show robustness to bandwidth specification, we first calculate the mean-squared error (MSE)-minimizing bandwidth (Calonico et al., 2014) for each year and outcome, and estimate results using the median of the yearly optimal bandwidths. These estimates are included in Appendix Figure A5, and with 95% confidence intervals in Appendix Figure A6. Results remain comparable except for cropland in 2006 and 2010, where the smaller MSE-optimal bandwidth yields significant discontinuities. In those years, however, Appendix Figure A7 is presented to compare models to underlying data, which shows no visible jump at the border.

We then estimate discontinuities for all bandwidths between 5 km and 80 km at 5 km increments (Appendix Figures A8 and A9). Regardless of bandwidth, results are nearly all statistically indistinguishable from zero prior to 2011 and nearly all significant and negative in wartime years, with the high-rainfall exceptions (2019 and 2020 for cropland; 2019 for NDVI) also appearing across bandwidths. Figures A10 and A11 repeat this exercise with the linear model. Linear results are sensitive to bandwidth, particularly for cropland at longer bandwidths, but the low bandwidths recommended for linear specifications (Lee and Lemieux, 2010) reproduce our findings, and the negative wartime shift holds across bandwidths (see also Appendix Figure A13).

B. Difference-in-discontinuities regressions

We re-estimate Equation 2 for each bandwidth between 5 km and 80 km in 5 km increments (Appendix Figure A12), and a linear version of Equation 2 with 2nd order terms omitted and triangular weights equal to $\frac{bw - |d|}{bw}$, where bw is the bandwidth (Appendix Figure A13). Results are statistically significant and negative at all bandwidths for all outcomes, with little variation in magnitude and comparable estimates across linear and quadratic specifications.

We repeat both exercises for the models with precipitation interactions (Appendix Figures A14 and A15, corresponding to Columns 2, 4, and 6 of Table 1). Outside of very low bandwidths in the quadratic model for cropland as an outcome (Panel A in Figure A14), the coefficient *Syria* $\times$ *Wartime* is negative and significant at at least the 5-percent level in every specification and bandwidth, while *Syria* $\times$ *Wartime* $\times$ *Precipitation* is significant and positive. This supports our findings of wartime declines in Syria along both margins, alongside increased vulnerability to variable precipitation.

6 Conclusion

This investigation contributes to the literature at the intersection of conflict, climate, and agriculture by providing quasi-experimental evidence that prolonged conflict increases agricultural vulnerability to climate shocks through the destruction of irrigation infrastructure. While a substantial body of work has established causal links between climate shocks and conflict risk, our study investigates the causal pathway linking conflict to increased climate vulnerability, which has received comparatively little rigorous empirical attention. By exploiting the geographic discontinuity at the Syria–Türkiye border, our design isolates the effect of conflict from confounding regional trends, offering credibly causal estimates of conflict-induced climate vulnerability that prior work has not been able to provide.

Negligible spatial discontinuities at the Syria-Türkiye border along intensive and extensive margins prior to the civil war, alongside large discontinuities during the war, are suggestive of a large negative effect of the war on agricultural land use and vegetation in Syria. We argue, given larger observed spatial discontinuities in agricultural outcomes in years of poor precipitation after the onset of the civil war, combined with evidence of large negative spatial discontinuities in evapotranspiration in such low-rainfall years, that degradation and destruction of irrigation infrastructure left Syrian agriculture without proper access to water and exposed to fluctuations in

rainfall.

The main policy issue which remains to be seen is whether, now that hostilities have ceased, abandoned Syrian cropland will return to its extent before the war and whether crop greenness will recover. Recent news gives some reason for optimism. In 2022, FAO and UK Aid announced the Building Local Resilience in Syria (BLRS) project, a £15.3M program which plans to rehabilitate irrigation and drainage network to provide access to water for 21,000 Syrian households (FAO, 2022). Likewise, the World Bank Group (WBG) has begun to re-engage with Syria, with an initial focus on restoring essential services and social stability through basic infrastructure and utilities. In 2026 the WBG announced a new US$150M water sector project to help Syria restore water supply services and secure reliable access to water. Additionally, recent reports indicate increasing efforts by the new government to combat the expansion of illegal unlicensed wells which proliferated during the conflict (Syrian Arab News Agency, 2026).

Whether such efforts to rebuild irrigation infrastructure and water governance are successful is likely to considerably influence the future trajectory of Syrian agriculture. Assuming our findings of increased vulnerability to poor rainfall hold in the absence of a broad effort to rebuild irrigation systems, the intensifying impacts of climate change are likely to increase precisely the drought conditions to which Syria has demonstrated increased vulnerability (Kelley et al., 2015). As such, concerted efforts to rebuild agricultural infrastructure should be among the top priorities for Syrian policymakers, humanitarian donors, and the international community as Syria begins its post-war reconstruction.

Several avenues for continued research emerge from this study. First, research which supports the main policy question, identifying the return of abandoned cropland following reconstruction efforts, will offer a natural test of whether the vulnerability documented here is reversible through infrastructure rehabilitation or whether the damage left by conflict is more persistent. Second, a central limitation of this study is the near-absence of time-series data on the location, type, and condition of irrigation infrastructure in Syria. We attempt to support our interpretation through the evapotranspiration analysis, but direct measurement of infrastructure destruction remains unavailable. Advances in the detection of irrigation infrastructure would allow future work to move from inferring the irrigation mechanism to directly estimating it. Lastly, applying the geographic regression discontinuity approach used here to other conflict borders where irrigation-dependent agriculture is present would help assess the generalizability of these findings beyond the Syrian case.

7 Appendix

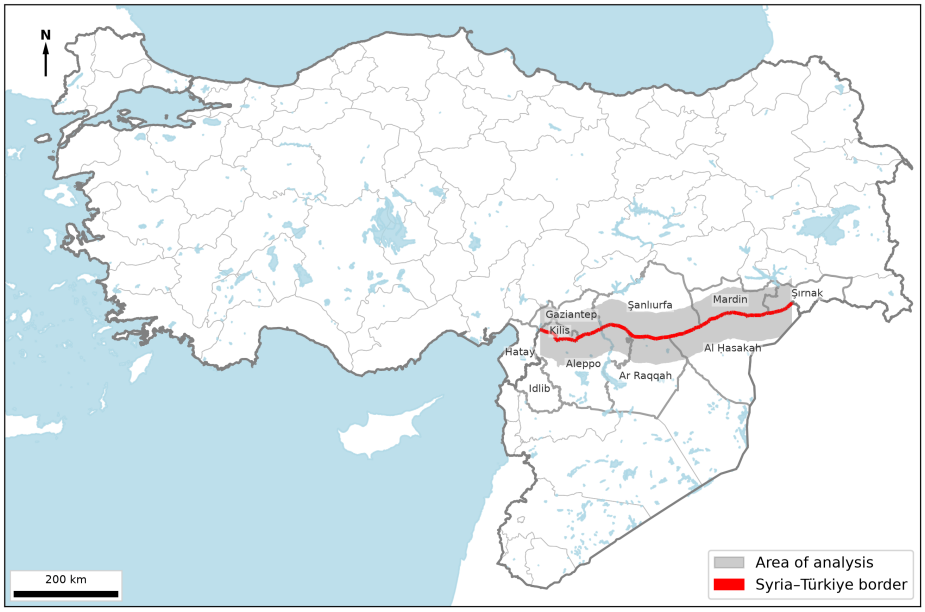


Figure A1: Area of analysis

Sources: World Bank Development Data Hub; and Digital Charts of the World. Note: Border provinces and governorates have been highlighted and labeled for context. Country shapefiles are the Official Boundaries shapefiles from the World Bank Development Data Hub. Inland water shapefiles are sourced from Digital Charts of the World.

Table A1: Summary statistics

	Cropland		NDVI		Precipitation		ET	
Year	SYR	TUR	SYR	TUR	SYR	TUR	SYR	TUR
2001	0.755	0.459	0.524	0.519	2.287	2.894	1.359	1.122
2002	0.710	0.447	0.403	0.429	1.622	2.135	1.017	1.039
2003	0.715	0.443	0.457	0.448	2.057	2.867	1.186	1.140
2004	0.713	0.443	0.470	0.481	1.572	2.038	1.130	0.992
2005	0.691	0.432	0.418	0.460	1.232	1.707	1.097	1.089
2006	0.672	0.418	0.415	0.453	1.813	2.338	1.165	1.117
2007	0.656	0.422	0.420	0.470	1.745	2.244	1.022	1.027
2008	0.587	0.346	0.275	0.374	0.893	1.242	0.584	0.715
2009	0.583	0.384	0.451	0.562	1.511	2.100	0.992	1.094
2010	0.659	0.422	0.528	0.559	1.329	1.856	1.200	1.160
2011	0.605	0.407	0.370	0.481	1.690	2.255	0.861	1.036
2012	0.559	0.410	0.401	0.447	1.369	1.867	0.973	1.091
2013	0.638	0.449	0.510	0.575	1.387	1.874	1.238	1.301
2014	0.635	0.429	0.399	0.490	1.375	1.795	0.928	1.003
2015	0.685	0.470	0.483	0.584	1.770	2.379	1.238	1.322
2016	0.621	0.453	0.407	0.511	1.322	1.645	0.964	1.176
2017	0.533	0.440	0.375	0.473	1.511	1.837	0.902	1.068
2018	0.521	0.442	0.340	0.494	1.599	2.008	0.777	1.076
2019	0.731	0.467	0.606	0.496	2.714	3.371	1.441	1.263
2020	0.719	0.464	0.497	0.603	2.227	2.811	1.346	1.383
2021	0.556	0.443	0.319	0.559	1.000	1.297	0.710	1.078
2022	0.311	0.415	0.352	0.443	1.473	1.861	0.772	1.008
2023	0.454	0.439	0.466	0.540	1.571	2.120	1.120	1.260

Notes: Above are the averages for variables of interest for each year for the entire area of analysis (within 50km of the border), separately for Syria and Türkiye. Cropland is a dummy variable, equal to one for land that was cropped in year t, and zero otherwise. The average of this dummy variable represents the percent of pixels which were cropped. NDVI is an index which measures greenness which can vary between negative and positive one, where an NDVI of one equals maximum greenness. NDVI values above represent the average NDVI value of cropland in each country (i.e. non-Cropland pixels were omitted from estimates). Both precipitation and evapotransiration are measured in units of mm/day and represent averages during the cropping period (Feb-Apr).

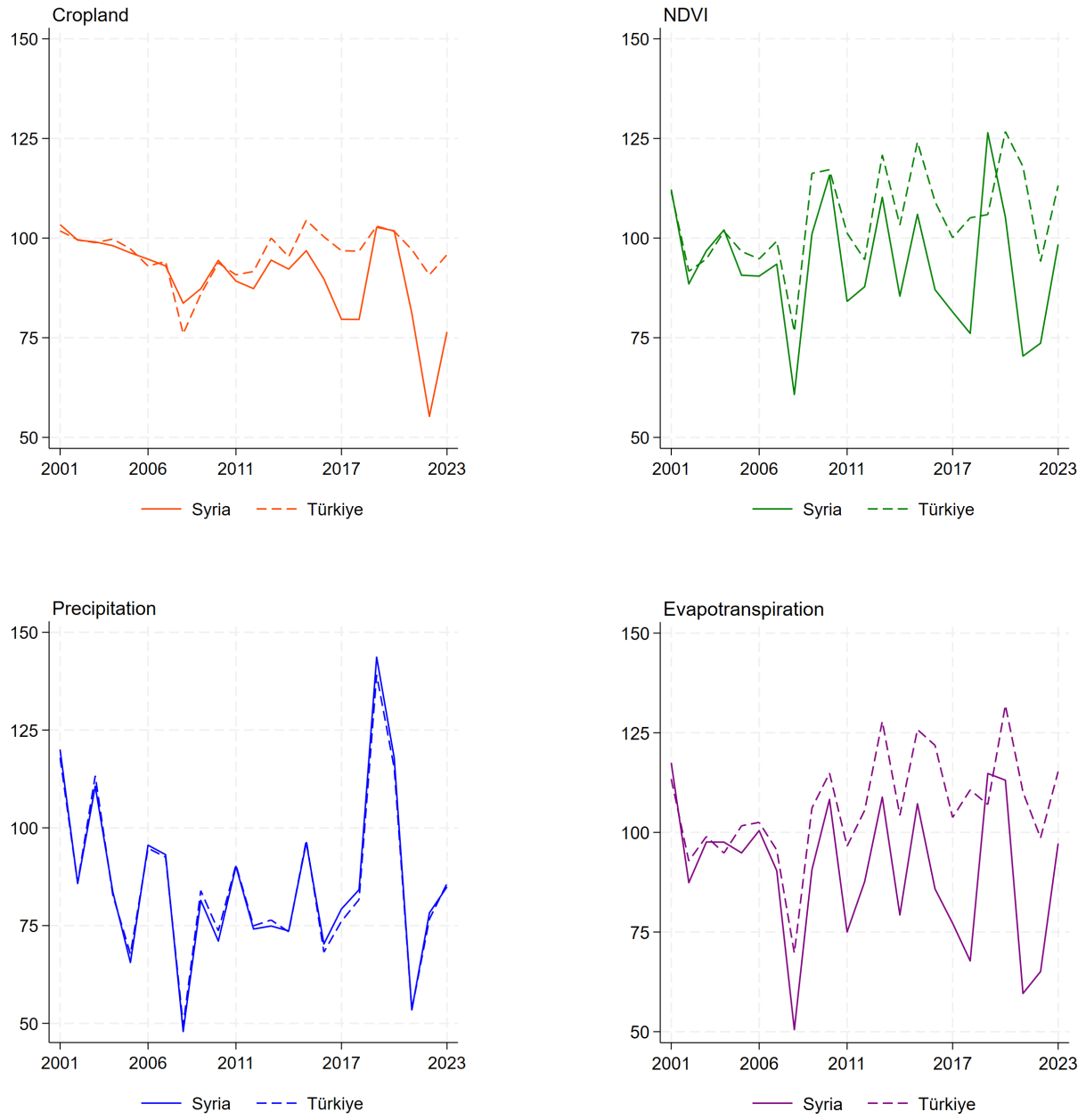


Figure A2: Averages of key variables by year, indexed

Note: Values above represent averages of relevant variables for the territory within 25 km of the Syria-Türkiye border, indexed to the country-specific averages for the years between 2001 and 2004, with 100 set as the baseline. NDVI figures represent averages among only data points which are cropland. ET figures represent averages among pixels which were classified as cropland in at least 80% of years pre-conflict.

Table A2: Cropland Results by Year

	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011
$Syria$	0.023	-0.009	-0.011	-0.029	-0.021	-0.010	-0.013	0.027	-0.004	-0.036	-0.067**
	(0.027)	(0.029)	(0.030)	(0.029)	(0.028)	(0.029)	(0.029)	(0.029)	(0.029)	(0.029)	(0.029)
d	0.026***	0.027***	0.027***	0.027***	0.026***	0.027***	0.025***	0.021***	0.026***	0.028***	0.029***
	(0.003)	(0.003)	(0.003)	(0.002)	(0.003)	(0.003)	(0.003)	(0.002)	(0.002)	(0.003)	(0.003)
$Syria$ x d	-0.031***	-0.032***	-0.033***	-0.032***	-0.030***	-0.031***	-0.030***	-0.020***	-0.029***	-0.030***	-0.030***
	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)
d^2	0.000***	0.000***	0.000***	0.000***	0.000***	0.000***	0.000***	0.000***	0.000***	0.000***	0.000***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
$Syria$ x d^2	-0.000***	-0.000***	-0.000***	-0.000***	-0.000***	-0.000***	-0.000***	-0.000***	-0.000***	-0.000***	-0.001***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Constant	0.860***	0.863***	0.900***	0.862***	0.798***	0.818***	0.776***	0.712***	0.917***	0.964***	0.893***
	(0.024)	(0.026)	(0.026)	(0.026)	(0.025)	(0.026)	(0.025)	(0.025)	(0.026)	(0.025)	(0.026)
Observations	244,958	244,958	244,958	244,958	244,958	244,958	244,958	244,958	244,958	244,958	244,958
R^2	0.295	0.288	0.288	0.287	0.281	0.280	0.274	0.263	0.286	0.276	0.274

Notes: Each column presents coefficient estimates for separate year-specific regressions. All regression results are estimated using Equation 1. The *Syria* coefficient above is denoted as β_1 in Equation 1. The *d* term represents α_1, the *Syria x d* term represents α_2, the d^2 term represents α_3, and the *Syria x* d^2 term represents α_4.

Table A2: Cropland Results by Year (continued)

	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023
$Syria$	-0.054*	-0.065**	-0.068**	-0.084***	-0.092***	-0.131***	-0.115***	-0.027	-0.013	-0.129***	-0.220***	-0.074**
	(0.031)	(0.029)	(0.030)	(0.029)	(0.031)	(0.033)	(0.032)	(0.027)	(0.027)	(0.032)	(0.035)	(0.033)
d	0.029***	0.028***	0.030***	0.029***	0.029***	0.029***	0.028***	0.028***	0.027***	0.028***	0.028***	0.027***
	(0.003)	(0.002)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)
$Syria$ x d	-0.034***	-0.030***	-0.033***	-0.031***	-0.034***	-0.036***	-0.036***	-0.030***	-0.031***	-0.034***	-0.043***	-0.042***
	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)
d^2	0.000***	0.000***	0.000***	0.000***	0.000***	0.000***	0.000***	0.000***	0.000***	0.000***	0.000***	0.000***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
$Syria$ x d^2	-0.000***	-0.000***	-0.000***	-0.000***	-0.000***	-0.000***	-0.000***	-0.000***	-0.000***	-0.000***	-0.000***	-0.000***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Constant	0.946***	0.986***	0.921***	0.983***	0.917***	0.870***	1.004***	1.007***	0.920***	0.881***	0.956***	0.898***
	(0.028)	(0.026)	(0.027)	(0.027)	(0.027)	(0.027)	(0.028)	(0.026)	(0.026)	(0.028)	(0.028)	(0.029)
Observations	244,958	244,958	244,958	244,958	244,958	244,958	244,958	244,958	244,958	244,958	244,958	244,958
R^2	0.268	0.260	0.257	0.257	0.252	0.247	0.252	0.276	0.268	0.220	0.249	0.241

Notes: Each column presents coefficient estimates for separate year-specific regressions. All regression results are estimated using Equation 1. The *Syria* coefficient above is denoted as β_1 in Equation 1. The *d* term represents α_1, the *Syria x d* term represents α_2, the d^2 term represents α_3, and the *Syria x* d^2 term represents α_4.

Table A3: NDVI Results by Year

	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011
$Syria$	-0.000	-0.004	0.011	-0.002	-0.005	0.012**	0.007	-0.027***	0.000	-0.005	-0.039***
	(0.008)	(0.008)	(0.006)	(0.007)	(0.005)	(0.006)	(0.007)	(0.008)	(0.008)	(0.006)	(0.009)
d	0.001*	0.002**	0.001	0.001	-0.001*	0.000	-0.001	0.002***	0.001	0.001**	0.003***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
$Syria$ x d	-0.002*	-0.004***	-0.000	-0.000	0.000	-0.002***	-0.001	-0.005***	-0.008***	-0.001*	-0.008***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
d^2	0.000	0.000	0.000	0.000	-0.000**	0.000	-0.000	0.000***	0.000**	0.000***	0.000**
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
$Syria$ x d^2	-0.000**	-0.000**	-0.000**	-0.000***	0.000	-0.000	-0.000	-0.000	-0.000	-0.000***	-0.000
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Constant	0.553***	0.497***	0.531***	0.498***	0.500***	0.577***	0.538***	0.530***	0.647***	0.556***	0.585***
	(0.008)	(0.007)	(0.007)	(0.007)	(0.006)	(0.005)	(0.007)	(0.009)	(0.008)	(0.006)	(0.009)
Observations	148,830	141,912	142,071	141,850	137,548	133,725	132,247	114,486	118,330	132,205	123,945
R^2	0.321	0.278	0.280	0.275	0.289	0.366	0.319	0.487	0.450	0.422	0.469

Notes: Each column presents coefficient estimates for separate year-specific regressions. All regression results are estimated using Equation 1. The *Syria* coefficient above is denoted as β_1 in Equation 1. The *d* term represents α_1, the *Syria x d* term represents α_2, the d^2 term represents α_3, and the *Syria x* d^2 term represents α_4.

Table A3: NDVI Results by Year (continued)

	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023
$Syria$	-0.012*	-0.034***	-0.104***	-0.087***	-0.120***	-0.107***	-0.113***	0.068***	-0.117***	-0.192***	-0.112***	-0.053***
	(0.007)	(0.008)	(0.008)	(0.008)	(0.008)	(0.007)	(0.010)	(0.009)	(0.007)	(0.010)	(0.009)	(0.009)
d	0.003***	0.001	0.004***	0.005***	0.005***	0.003***	0.003***	0.001	0.004***	0.001	0.003***	0.002***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
$Syria$ x d	-0.007***	-0.003***	-0.005***	-0.008***	-0.007***	-0.004***	-0.008***	0.001*	-0.005***	-0.006***	-0.005***	-0.005***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
d^2	0.000***	0.000	0.000***	0.000***	0.000***	0.000***	0.000*	0.000	0.000***	0.000	0.000*	0.000**
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
$Syria$ x d^2	-0.000	-0.000	-0.000***	-0.000***	-0.000***	-0.000***	-0.000	-0.000***	-0.000***	0.000*	0.000	-0.000
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Constant	0.518***	0.600***	0.537***	0.672***	0.604***	0.600***	0.719***	0.501***	0.598***	0.678***	0.555***	0.628***
	(0.007)	(0.008)	(0.008)	(0.008)	(0.007)	(0.008)	(0.009)	(0.008)	(0.006)	(0.008)	(0.008)	(0.008)
Observations	118,642	132,686	130,112	141,521	131,357	119,044	117,770	146,954	145,030	122,132	88,673	109,420
R^2	0.352	0.309	0.539	0.379	0.613	0.442	0.578	0.437	0.373	0.593	0.438	0.289

Each column presents coefficient estimates for separate year-specific regressions. All regression results are estimated using Equation 1. The *Syria* coefficient above is denoted as β_1 in Equation 1. The *d* term represents α_1, the *Syria x d* term represents α_2, the d^2 term represents α_3, and the *Syria x* d^2 term represents α_4.



(a) 2010 (b) 2011 (c) 2012 (d) 2013 (e) 2014

Figure A3: Satellite photography on either side of Syria-Türkiye border

Note: The images above are derived from Google Earth satellite imagery and show the area near 36°40′N, 39°40′E in December of each year. North is oriented upward; areas above the white line are in Türkiye, and areas below are in Syria.

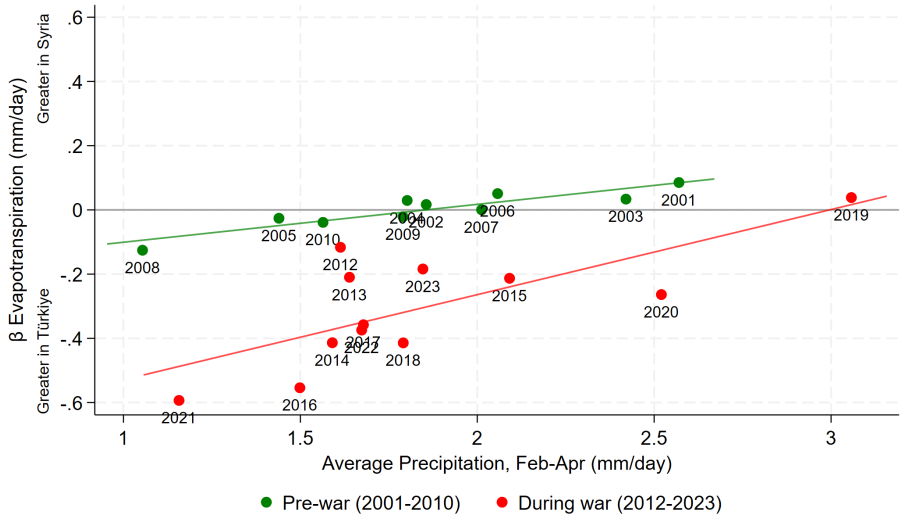
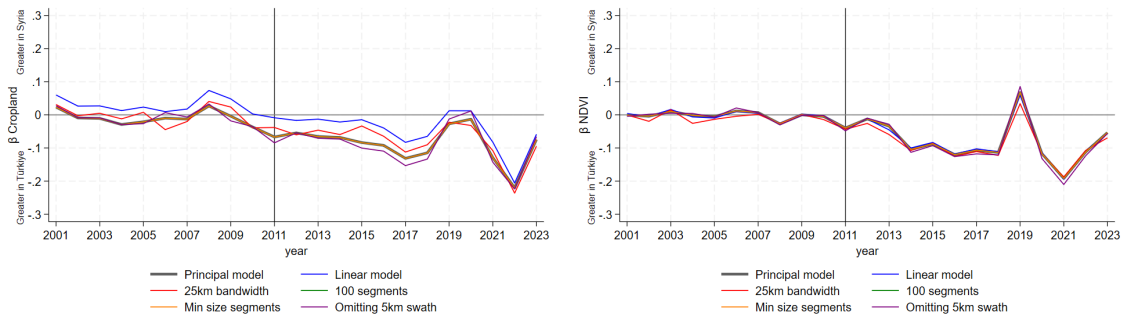


Figure A4: Discontinuities in evapotranspiration by average annual precipitation

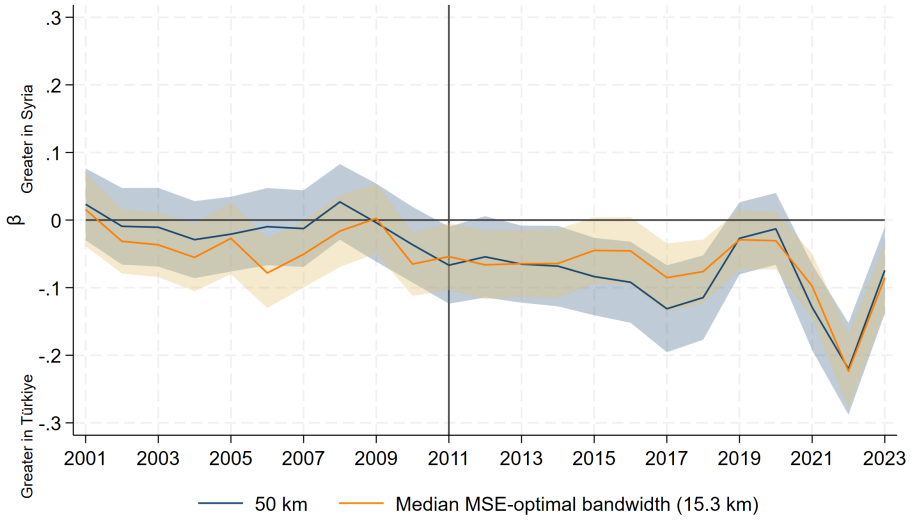
Note: Vertical axis values represent discontinuity coefficients from regression estimates using Equation 1. Horizontal axis values represent average precipitation within 20 km of the Syria-Türkiye border in year t. Fit lines represent linear fit lines by time period. 2011 cannot be assigned to either pre-war or during-war status because the conflict started during the year's cropping season and is therefore omitted.



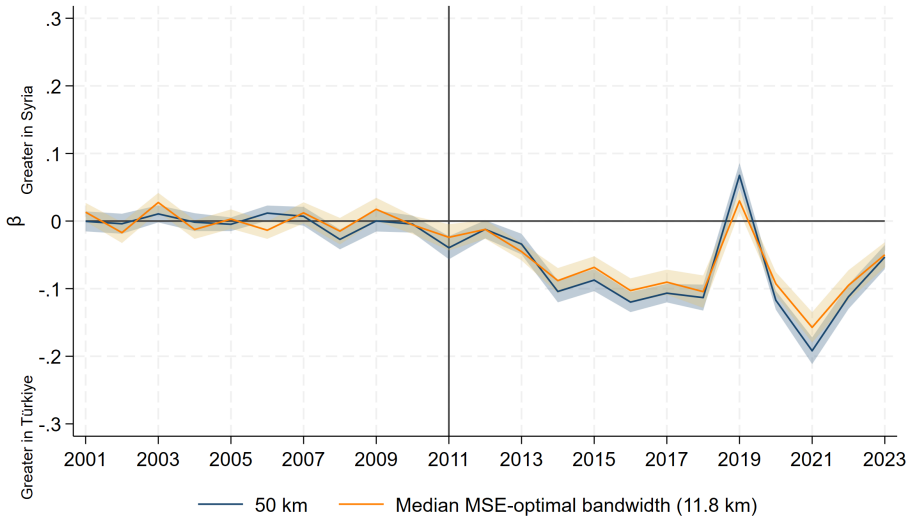
(a) Cropland (b) NDVI

Figure A5: Robustness checks of year-specific regressions

Note: *Principal model* is the model used throughout this investigation, as outlined in Equation 1 and as outlined in Tables A2 and A3. *Linear model* omits quadratic terms from the principal regression equation, uses a 25 km bandwidth, and uses statistical weights equal to the inverse of 25 km minus the absolute distance between pixel i and the Syria-Türkiye border. *25 km bandwidth* represents estimates using the principal Equation 1, while only including pixels within 25 km of the Syria-Türkiye border in analysis. *100 segments* resizes the segments used in fixed effects, such that the border is divided into 100 equal sized segments (as opposed to the 250 which are typically used), while *Min size segments* reduces segments to the minimum possible width given spatial data resolution (500m). *Omitting 5 km swath* omits pixels within 2.5 km of the border on either side in analysis, reducing the risk of outliers near the border from having an outsized effect.



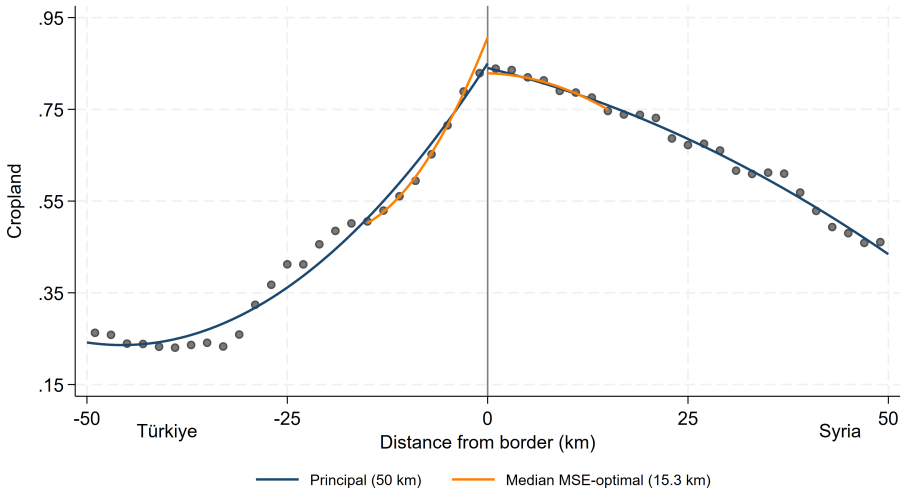
(a) Cropland



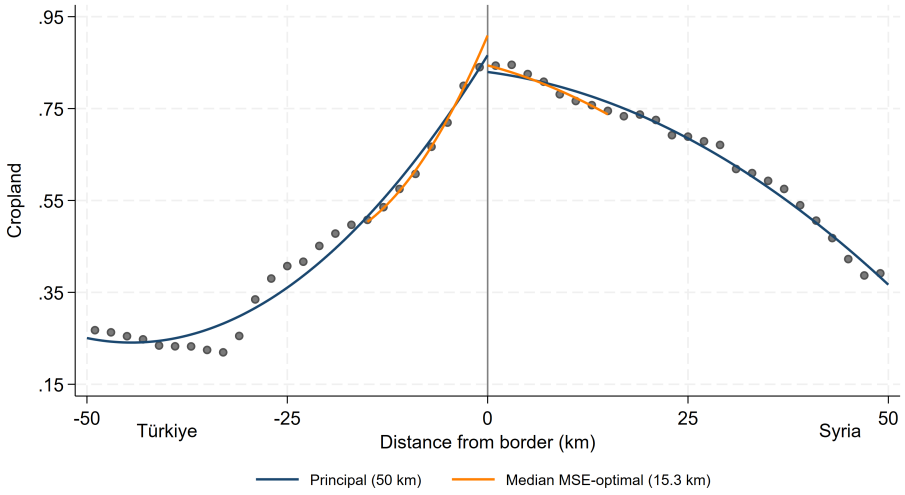
(b) NDVI

Figure A6: Regression estimates by year, MSE-optimal and principal model bandwidths

Note: Regression results represent estimates of the GRD discontinuity coefficient (β_1) using Equation 1. Error bars represent 95 percent confidence intervals, clustering standard errors by vertical segment. All results presented were calculated using the same regression equation, but using alternate bandwidths. The principal bandwidth of 50 km is shown, alongside the median of the calculated mean-squared error optimizing year-specific bandwidths.



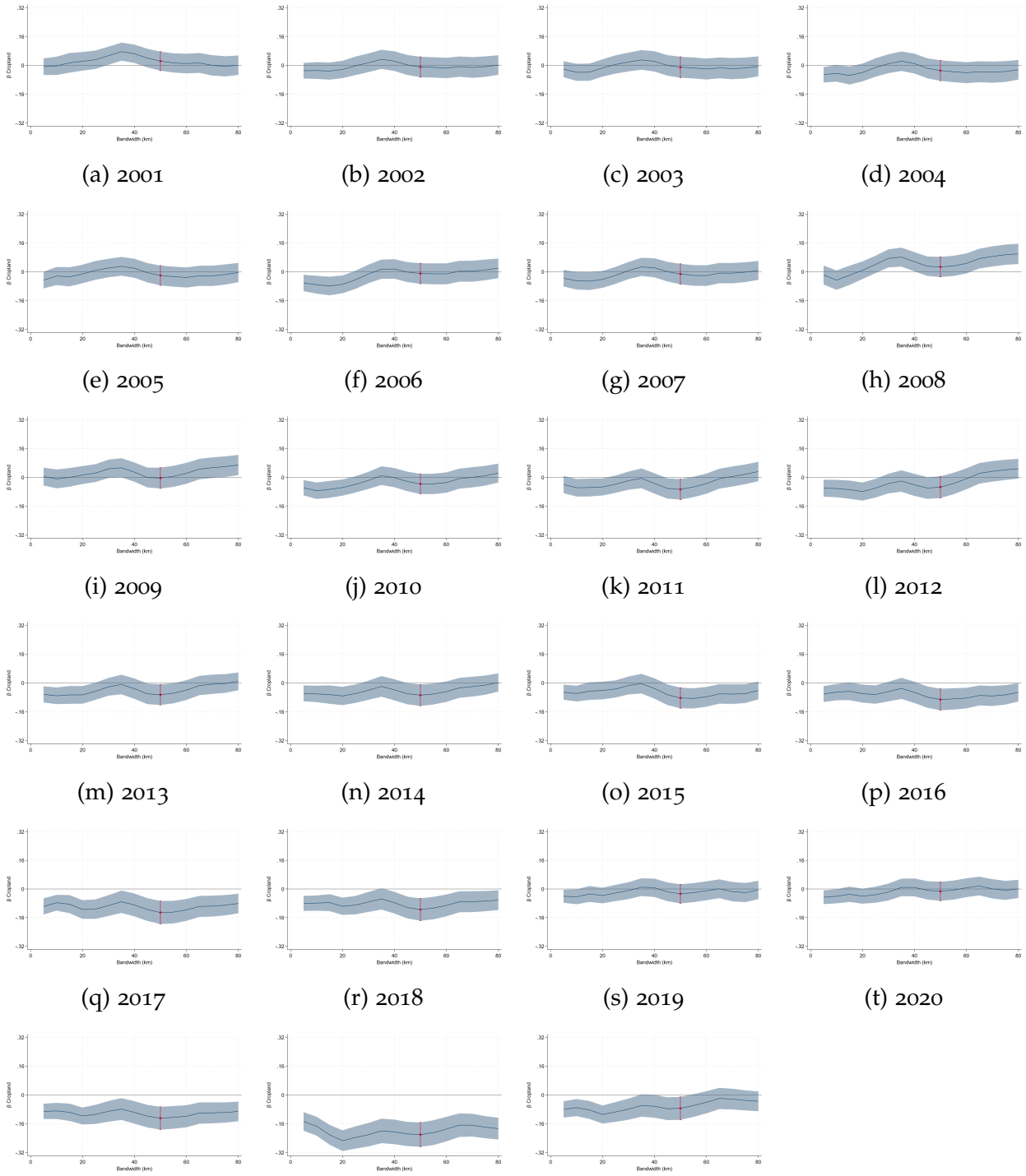
(a) 2006



(b) 2010

Figure A7: Cropland regression fit lines by distance to the border, MSE-optimal and principal model bandwidths

Note: Horizontal axis values indicate distance to the border (negative values for Türkiye and positive values for Syria). Plotted lines represent regression estimates while dots represent the averages of the underlying data by distance by 2 km bins. All results presented were calculated using Equation 1, with varying bandwidths. The principal bandwidth of 50 km is shown, alongside the median of the calculated mean-squared error optimizing year-specific bandwidths.



(a) 2001 (b) 2002 (c) 2003 (d) 2004

(e) 2005 (f) 2006 (g) 2007 (h) 2008

(i) 2009 (j) 2010 (k) 2011 (l) 2012

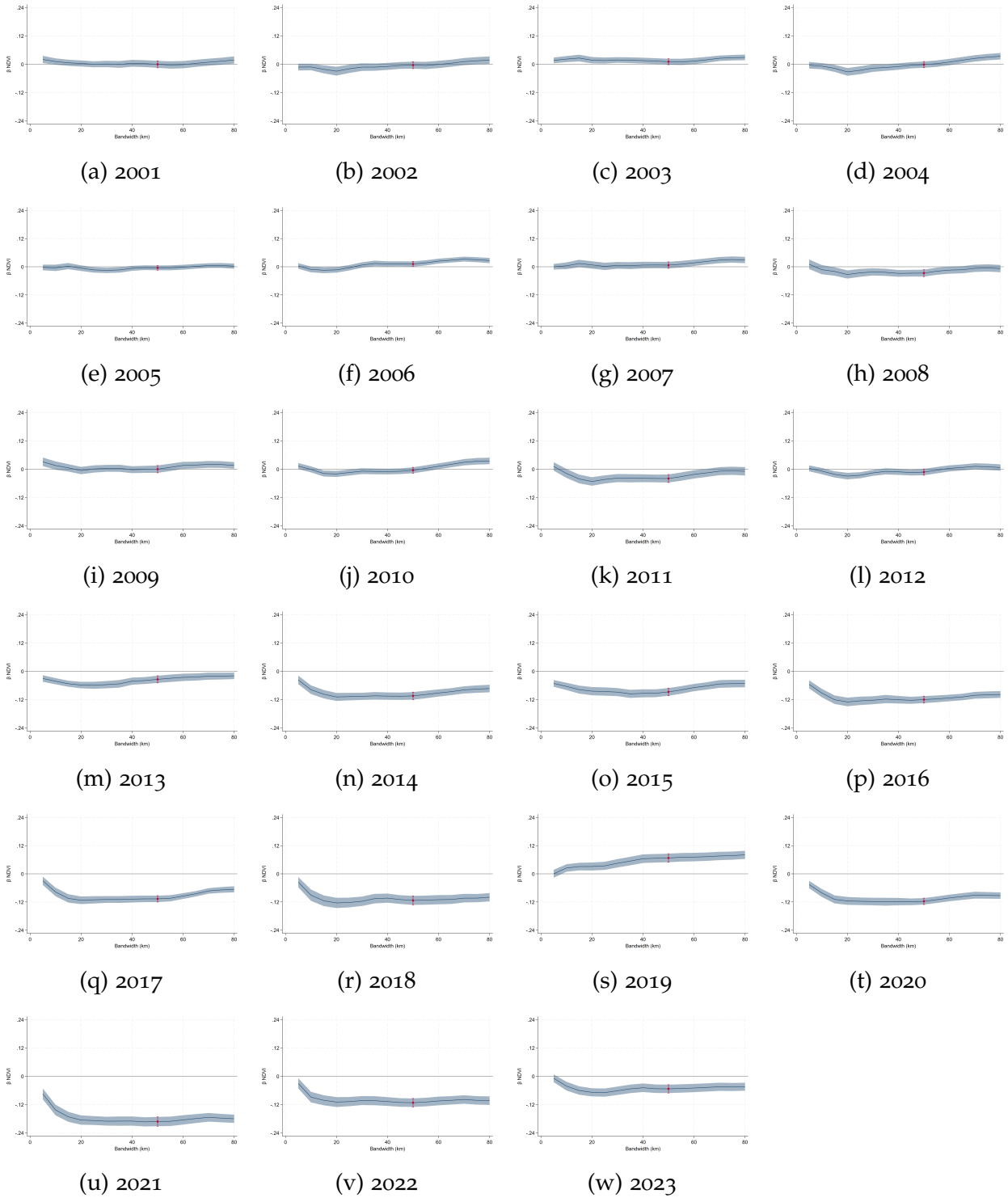
(m) 2013 (n) 2014 (o) 2015 (p) 2016

(q) 2017 (r) 2018 (s) 2019 (t) 2020

(u) 2021 (v) 2022 (w) 2023

Figure A8: Year-specific discontinuities in cropland by bandwidth using quadratic specification

Note: Vertical axis values represent GRD coefficient values using the principal quadratic regression specification (β_1 in Equation 1). The dark line represents coefficient estimates with surrounding shaded areas representing 95% confidence intervals. Horizontal axis values indicate the RDD bandwidth, or the distance north or south from the Syria/Türkiye border past which a data point will be excluded from analysis. Presented are bandwidths between 5 km and 80 km, in increments of 5 km. The values and error bars presented in red reflect estimates using the 50 km bandwidth employed throughout this analysis, as presented in Figure 3 and Appendix Table A2.



(a) 2001 (b) 2002 (c) 2003 (d) 2004

(e) 2005 (f) 2006 (g) 2007 (h) 2008

(i) 2009 (j) 2010 (k) 2011 (l) 2012

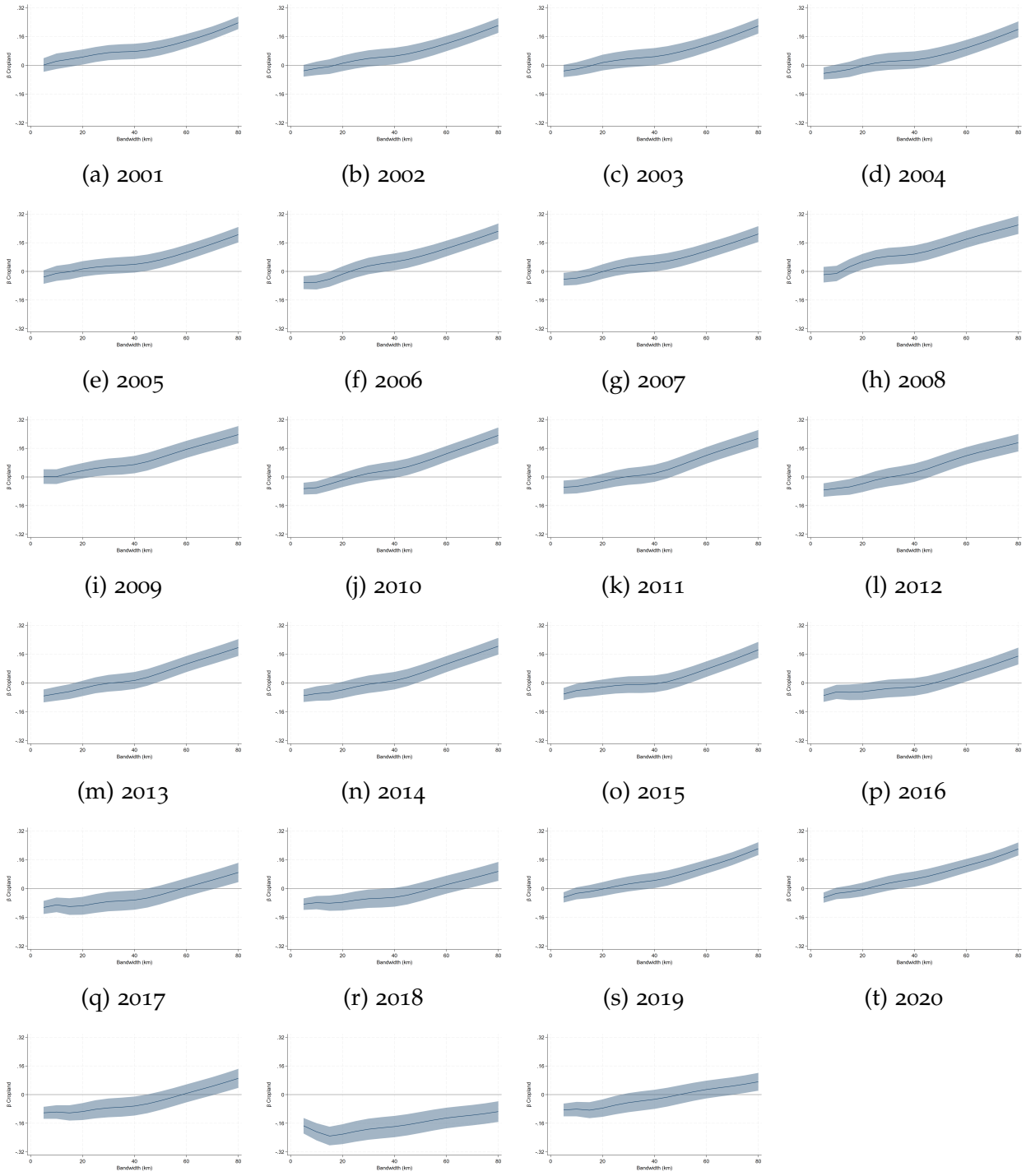
(m) 2013 (n) 2014 (o) 2015 (p) 2016

(q) 2017 (r) 2018 (s) 2019 (t) 2020

(u) 2021 (v) 2022 (w) 2023

Figure A9: Year-specific discontinuities in NDVI by bandwidth using quadratic specification

Note: Vertical axis values represent GRD coefficient values using the principal quadratic regression specification (β_1 in Equation 1). The dark line represents coefficient estimates with surrounding shaded areas representing 95% confidence intervals. Horizontal axis values indicate the RDD bandwidth, or the distance north or south from the Syria/Türkiye border past which a data point will be excluded from analysis. Presented are bandwidths between 5 km and 80 km, in increments of 5 km. The values and error bars presented in red reflect estimates using the 50 km bandwidth employed throughout this analysis, as presented in Figure 3 and Appendix Table A3.



(a) 2001 (b) 2002 (c) 2003 (d) 2004

(e) 2005 (f) 2006 (g) 2007 (h) 2008

(i) 2009 (j) 2010 (k) 2011 (l) 2012

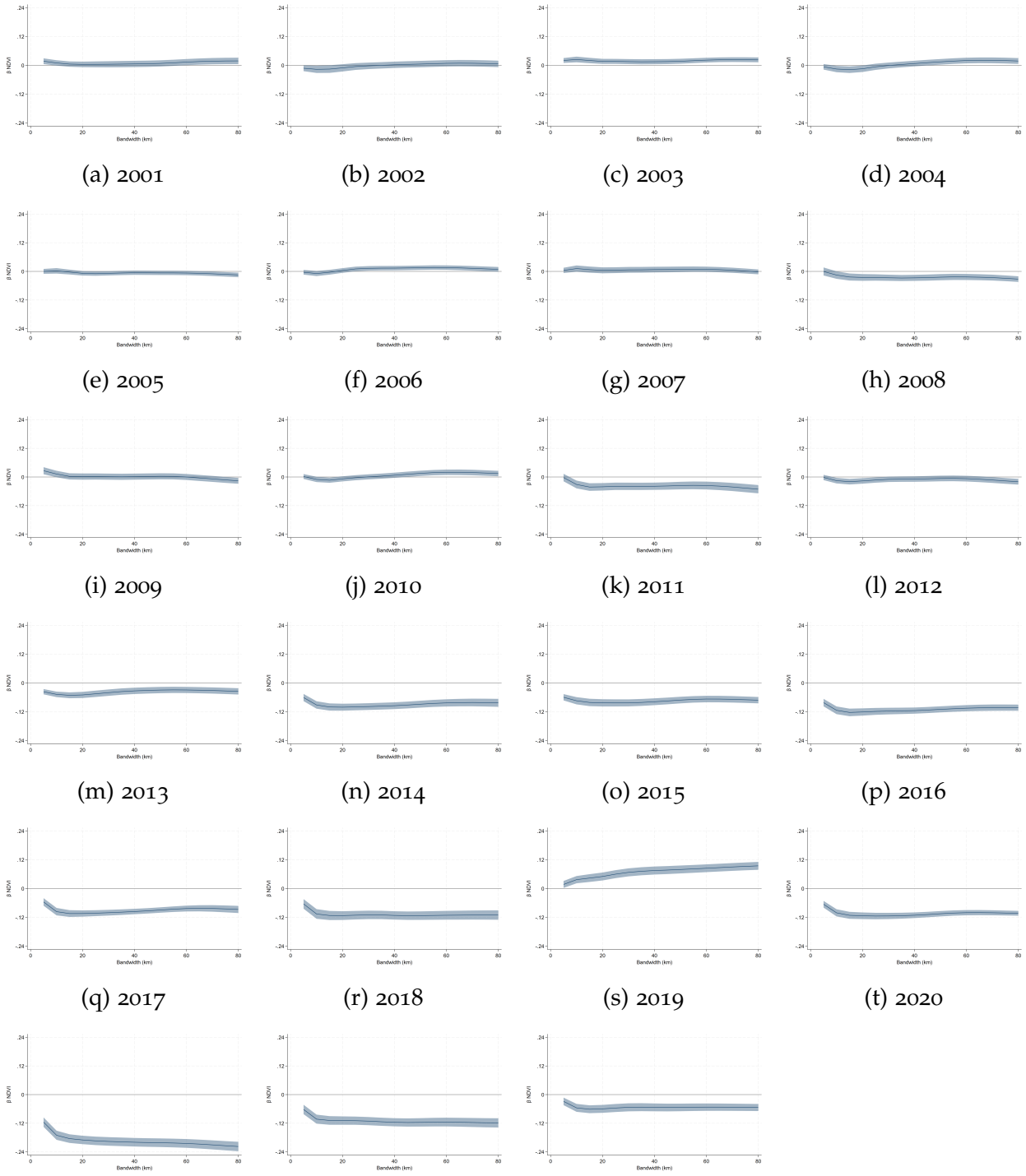
(m) 2013 (n) 2014 (o) 2015 (p) 2016

(q) 2017 (r) 2018 (s) 2019 (t) 2020

(u) 2021 (v) 2022 (w) 2023

Figure A10: Year-specific discontinuities in cropland by bandwidth using linear specification

Note: Vertical axis values represent GRD coefficient values using the linear regression specification (β_1 in Equation 4). The dark line represents coefficient estimates with surrounding shaded areas representing 95% confidence intervals. Horizontal axis values indicate the RDD bandwidth, or the distance north or south from the Syria/Türkiye border past which a data point will be excluded from analysis. Presented are bandwidths between 5 km and 80 km, in increments of 5 km.



(a) 2001 (b) 2002 (c) 2003 (d) 2004

(e) 2005 (f) 2006 (g) 2007 (h) 2008

(i) 2009 (j) 2010 (k) 2011 (l) 2012

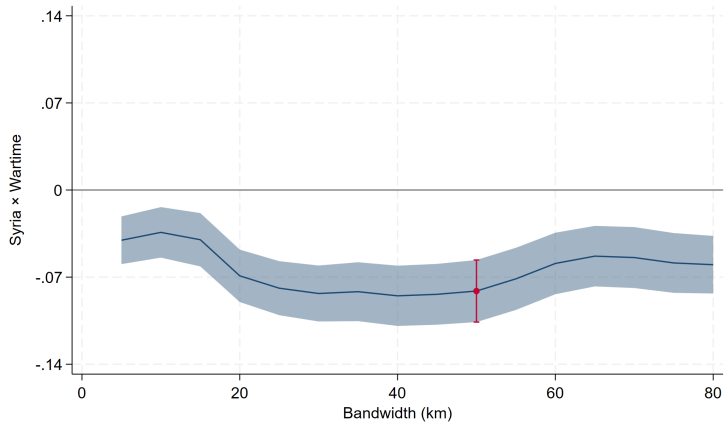
(m) 2013 (n) 2014 (o) 2015 (p) 2016

(q) 2017 (r) 2018 (s) 2019 (t) 2020

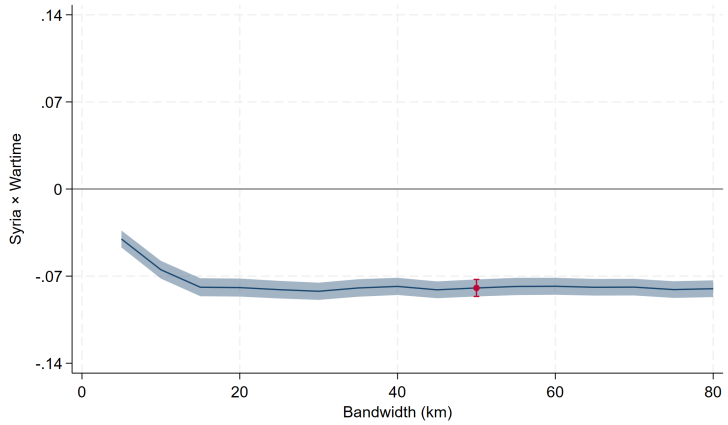
(u) 2021 (v) 2022 (w) 2023

Figure A11: Year-specific discontinuities in NDVI by bandwidth using linear specification

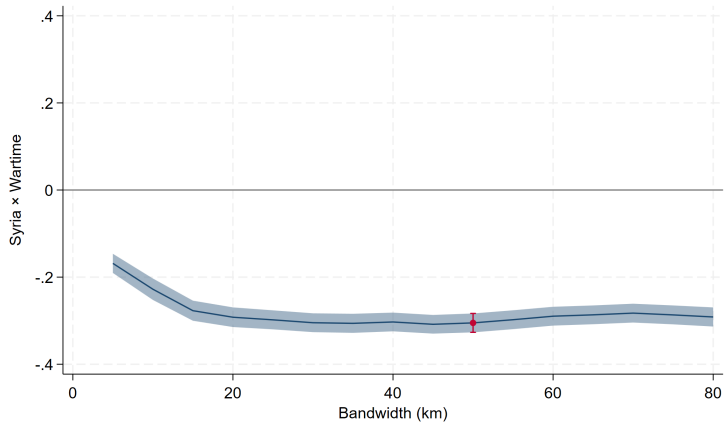
Note: Vertical axis values represent GRD coefficient values using the linear regression specification (β_1 in Equation 4). The dark line represents coefficient estimates with surrounding shaded areas representing 95% confidence intervals. Horizontal axis values indicate the RDD bandwidth, or the distance north or south from the Syria/Türkiye border past which a data point will be excluded from analysis. Presented are bandwidths between 5 km and 80 km, in increments of 5 km.



(a) Cropland



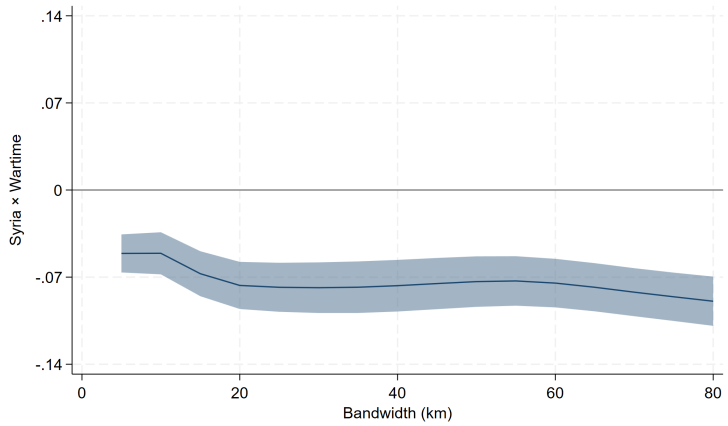
(b) NDVI



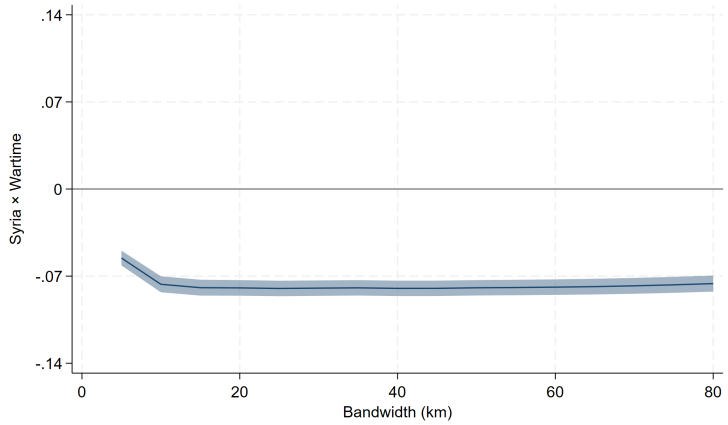
(c) Evapotranspiration

Figure A12: Difference-in-discontinuities coefficient using quadratic regression equation

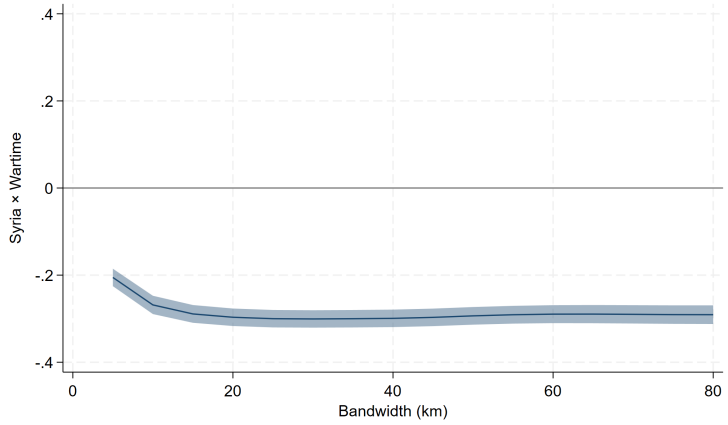
Note: Vertical axis values represent difference-in-discontinuities coefficient values using the principal quadratic regression specification (β_2 in Equation 2). The dark line represents coefficient estimates with surrounding shaded areas representing 95% confidence intervals. Horizontal axis values indicate the RDD bandwidth, or the distance north or south from the Syria/Türkiye border past which a data point will be excluded from analysis. Presented are bandwidths between 5 km and 80 km, in increments of 5 km. The values and error bars presented in red reflect estimates using the 50 km bandwidth employed throughout this analysis, as presented in Table 1.



(a) Cropland



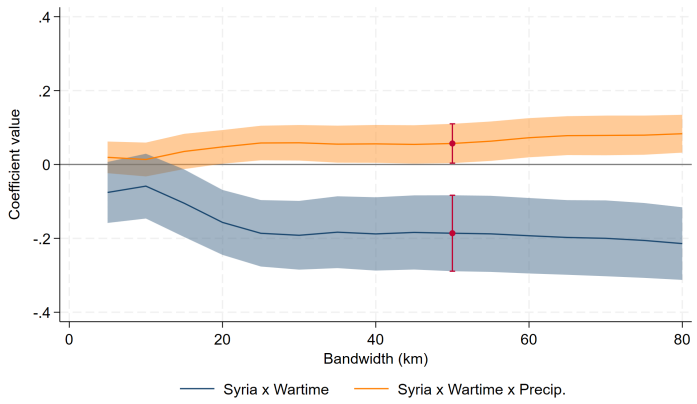
(b) NDVI



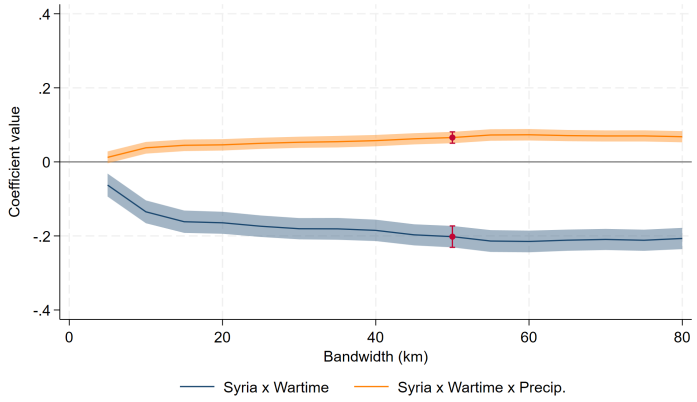
(c) Evapotranspiration

Figure A13: Difference-in-discontinuities coefficient using linear regression equation

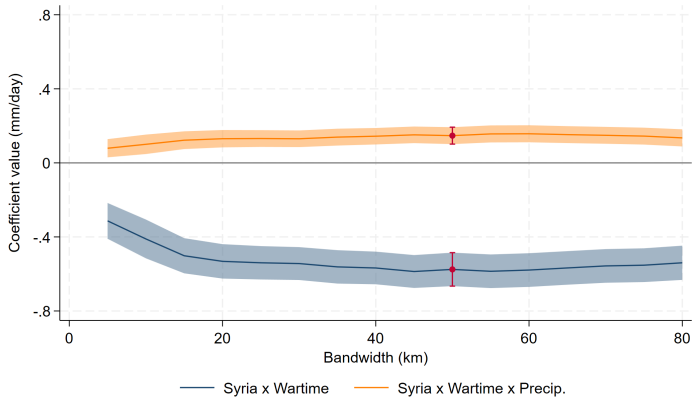
Note: Vertical axis values represent difference-in-discontinuities coefficient values using the linear regression specification (β_2 in Equation **??**). The dark line represents coefficient estimates with surrounding shaded areas representing 95% confidence intervals. Horizontal axis values indicate the RDD bandwidth, or the distance north or south from the Syria/Türkiye border past which a data point will be excluded from analysis. Presented are bandwidths between 5 km and 80 km, in increments of 5 km.



(a) Cropland



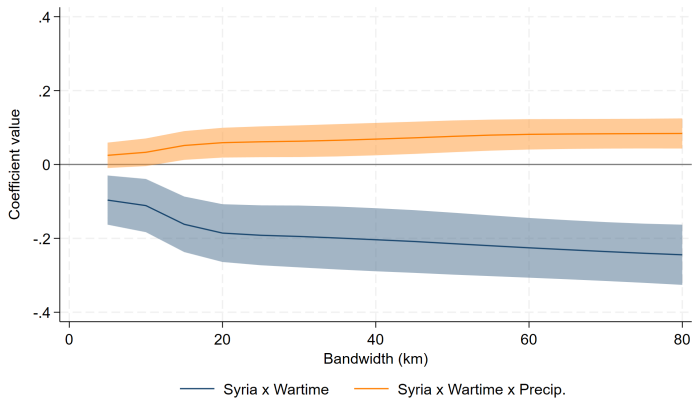
(b) NDVI



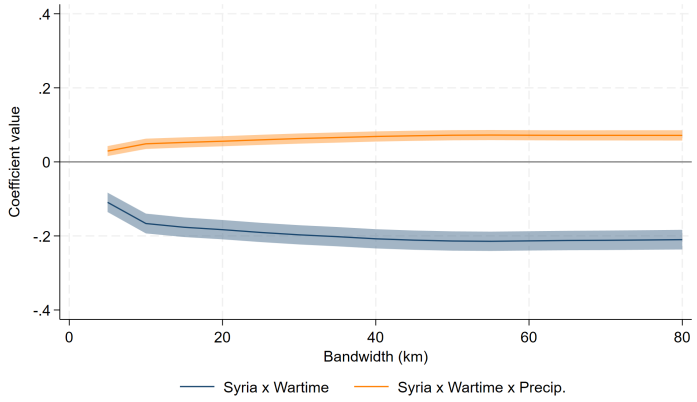
(c) Evapotranspiration

Figure A14: Difference-in-discontinuities coefficients using quadratic regression equation with precipitation interaction

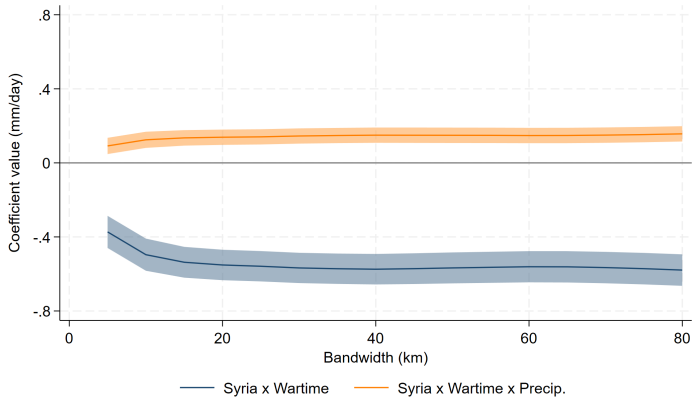
Note: Vertical axis values represent difference-in-discontinuities coefficient values using the quadratic regression specification with a precipitation interaction (Equation 3). The blue line represents difference-in-discontinuities coefficient estimates (β_2), and the orange line represents its interaction with precipitation (β_4). Shaded areas represent 95% confidence intervals. Horizontal axis values indicate the RDD bandwidth, or the distance north or south from the Syria/Türkiye border past which a data point will be excluded from analysis. Presented are bandwidths between 5 km and 80 km, in increments of 5 km. The values and error bars presented in red reflect estimates using the 50 km bandwidth employed throughout this analysis, as presented in Table 1.



(a) Cropland



(b) NDVI



(c) Evapotranspiration

Figure A15: Difference-in-discontinuities coefficients using linear regression equation with precipitation interaction

Note: Vertical axis values represent difference-in-discontinuities coefficient values using the linear regression specification with a precipitation interaction. The regression equation mirrors the quadratic specification (Equation 3), but with d^2 terms omitted. The blue line represents difference-in-discontinuities coefficient estimates (β_2), and the orange line represents its interaction with precipitation (β_4). Shaded areas represent 95% confidence intervals. Horizontal axis values indicate the RDD bandwidth, or the distance north or south from the Syria/Türkiye border past which a data point will be excluded from analysis. Presented are bandwidths between 5 km and 80 km, in increments of 5 km.

References

Atik, Omar, Anas Kadour, Ibrahim Mahmoud, Khalid Al Hasan et al. 2024. "Irrigation Water in Northwest Syria: Impact of the Recent Crisis and Drought." *Water*, 16(21): , p. 3101, DOI: http://dx.doi.org/10.3390/w16213101, License: CC BY 4.0.

Ayalew, Hailemariam, Guush Berhane, Meseret Wondale, and Clemens Breisinger. 2026. "Armed Conflict and Climate-Induced Weather Disruptions in Agricultural Input Use: Evidence From Ethiopia." *Agricultural Economics*, 57(1): , p. e70083, DOI: http://dx.doi.org/10.1111/agec.70083.

Brown, Jesslyn. 2018. "Climate and Conflict." *Annual Review of Economics*, 7(1): 577–617.

Burke, Marshall, Joel Ferguson, Solomon Hsiang, and Edward Miguel. 2024. "New Evidence on the Economics of Climate and Conflict." In *Handbook of the Economics of Conflict*. 1: Elsevier, 249–305, DOI: http://dx.doi.org/10.1016/bs.hoec.2024.10.008.

Calonico, Sebastian, Matias D. Cattaneo, and Rocío Titiunik. 2014. "Robust Nonparametric Confidence Intervals for Regression-Discontinuity Designs." *Econometrica*, 82(6): 2295–2326.

Corsi, Anna, and Harris Selod. 2023. *Land Matters: Can Better Governance and Management of Scarcity Prevent a Looming Crisis in the Middle East and North Africa?*. Washington, DC: World Bank.

Dell, M. 2010. "The Persistent Effects of Peru's Mining Mita." *Econometrica*, 78(6): 1863–1903.

Didan, Kamel. 2015. "MOD13A1 MODIS/Terra Vegetation Indices 16-Day L3 Global 500m SIN Grid V006." *NASA EOSDIS Land Processes DAAC*, 10.

FAO. 2017. "Counting the Cost: Agriculture in Syria after Six Years of Crisis."Technical report, Food and Agriculture Organization of the United Nations, Rome.

FAO. 2022. "Building Local Resilience in the Syrian Arab Republic (BLRS)." Project Profile GCP/SYR/031/UK, FAO, Rome, Italy.

FAO, and WFP. 2016. "Special Report: FAO/WFP Crop and Food Security Assessment Mission to the Syrian Arab Republic."Technical report, Food and Agriculture Organization of the United Nations and World Food Programme, Rome.

Ferguson, Joel, and Oliver Kim. 2026. "What Started China's Miracle? The View from Outer Space." March, Working paper.

Gelman, Andrew, and Guido Imbens. 2019. "Why High-Order Polynomials Should Not Be Used in Regression Discontinuity Designs." *Journal of Business & Economic Statistics*, 37(3): 447–456.

Gleick, Peter H. 2014. "Water, Drought, Climate Change, and Conflict in Syria." *Weather, Climate, and Society*, 6(3): 331–340, DOI: http://dx.doi.org/10.1175/WCAS-D-13-00059.1.

Grembi, Veronica, Tommaso Nannicini, and Ugo Troiano. 2016. "Do Fiscal Rules Matter?" *American Economic Journal: Applied Economics*, 8(3): 1–30.

Guarnieri, Eleonora, and Helmut Rainer. 2021. "Colonialism and Female Empowerment: A Two-Sided Legacy." *Journal of Development Economics*.

Jensen, John R. 2018. *Introductory digital image processing: a remote sensing perspective.*.(Ed. 4): : Pearson Education Inc..

Kelley, Colin P., Shahrzad Mohtadi, Mark A. Cane, Richard Seager, and Yochanan Kushnir. 2015. "Climate Change in the Fertile Crescent and Implications of the Recent Syrian Drought." *Proceedings of the National Academy of Sciences*, 112(11): 3241–3246, DOI: http://dx.doi.org/10.1073/pnas.1421533112.

Lee, David S., and Thomas Lemieux. 2010. "Regression Discontinuity Designs in Economics." *Journal of Economic Literature*, 48, p. 281–355.

Li, Xi-Ya, Xi Li, Ziying Fan, Li Mi, Tarek Kandakji, Zhen Song, Deren Li, and Xiao-Peng Song. 2022. "Civil war hinders crop production and threatens food security in Syria." *Nature Food*, 3(1): 38–46, DOI: http://dx.doi.org/10.1038/s43016-021-00432-4.

Martin-Shields, Charles P., and Wolfgang Stojetz. 2019. "Food Security and Conflict: Empirical Challenges and Future Opportunities for Research and Policy Making on Food Security and Conflict." *World Development*, 119 150–164, DOI: http://dx.doi.org/10.1016/j.worlddev.2018.07.011.

McGuirk, Eoin, and Marshall Burke. 2020. "The economic origins of conflict in Africa." *Journal of Political Economy*, 128(10): 3940–3997.

Michalopoulos, Stelios, and Elias Papaioannou. 2014. "National Institutions and Subnational Development in Africa." *The Quarterly Journal of Economics*, 129(1): 151–213, DOI: http://dx.doi.org/10.1093/qje/qjt029.

Picchetti, Pedro, Cristine C. X. Pinto, and Stéphanie Tramontin Shinoki. 2026. "Difference-in-Discontinuities: Estimation, Inference and Validity Tests." URL: https://doi.org/10.48550/arXiv.2405.18531, arXiv:2405.18531v2 [econ.EM].

Rouse, JW, Rüdiger H Haas, John A Schell, Donald W Deering et al. 1974. "Monitoring vegetation systems in the Great Plains with ERTS." *NASA special publication*, 351(1974): , p. 309.

Sulla-Menashe, Damien, and Mark A Friedl. 2018. "User guide to collection 6 MODIS land cover (MCD12Q1 and MCD12C1) product." *USGS: Reston, VA, USA* 1–18.

Syrian Arab News Agency. 2026. "Thousands of Illegal Wells in Rural Damascus and Daraa; Solutions Under Implementation." Syrian Arab News Agency (SANA), January, URL: https://sana.sy/locals/2379121/.

Tintchev, Kalin, and Laura Jaramillo. 2024. "Hanging Out to Dry? Long-Term Macroeconomic Effects of Drought in Fragile and Conflict-Affected States." IMF Working Paper WP/24/106, International Monetary Fund, Washington, DC.

World Bank. 2022. "Syria Economic Monitor: Lost Generation of Syrians."Technical Report 172268, World Bank, Washington, DC.